

STAFFORD’S CONJECTURE ON CYCLICITY OF TORSION MODULES

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ABSTRACT. Let $A_n(k)$ be the Weyl algebra over a field of characteristic zero. We prove that every nonzero $d \in A_n(k)$ admits $F, R, S \in A_n(k)$ such that $1 = dR + FdS$. Thus Stafford’s Conjecture 3.8 holds. **For $n \geq 1$, the multiplier F may be chosen as x^N , after a linear symplectic change of variables, where N is the Bernstein degree of d .** The proof studies a cyclic quotient through its characteristic variety and excludes nonzero quotients by conormal geometry at infinity.

AI comment Review guide. Suggested changes. Red text is proposed wording, not an accepted correction. Blue strike-through preserves the original text. Accept or revise the proposal before removing its wrappers. **Comments and action.** Every box states its status: *Proposed change* awaits author acceptance; *Action required* needs a proof repair or drafting decision; *Information* requests no manuscript edit; *Optional refinement* may be omitted. A different verified proof does not close a gap in the printed proof. **Who reviews.** Johanna can edit wording and integrate agreed changes; Max reviews the whole paper—Lean correspondence, including exact matches, definitions and printed proof steps. He checks mathematical gaps before asking Johanna for a text repair. Authors decide the geometric proof route. Follow *Mathematical proof* links for the readable comparison; no Lean code is needed. Resolve a box only after its stated task is done, by deleting its \AIcomment command.

AI comment AUDIT-01. Information — no edit requested. Corrections, clarifications and mathematical differences from the formal proof remain marked as AI additions. Long formal proof comparisons and declaration links are collected in the separate Overleaf supplement `lean_proof_details.tex`. The human-readable draft shows the mathematical qualifications beside the relevant passages; the side-by-side audit retains the full technical correspondence.

1. INTRODUCTION

Let k be a field of characteristic zero, and let

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle$$

be the n th Weyl algebra, with $[\partial_i, x_j] = \delta_{ij}$ and $[\partial_i, \partial_j] = [x_i, x_j] = 0$.

Stafford conjectured in 1978 (Conjecture 3.8 of [31, p. 438]) that for every nonzero $d \in A_n(k)$, one can find $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$. Stafford notes that this immediately implies $A_n(k)/dA_n(k)$ is cyclic, i.e. generated by a single element. [7] reported the resulting question — whether

every right torsion module over $A_n(k)$ is cyclic—as open [Section 3, Conjecture 3.4]. We prove Stafford’s conjecture:

Stafford’s Conjecture 3.8 asks whether every nonzero $d \in A_n(k)$ admits $F \in A_n(k)$ with $A_n(k) = dA_n(k) + FdA_n(k)$ [31, p. 438]. Stafford identifies this with stable cyclicity for finitely generated torsion right Weyl-algebra modules [31, pp. 437–438]. Bellamy’s survey states the weaker ordinary-cyclicity question for finitely generated torsion $A_n(\mathbb{C})$ -modules as open [7, Section 3, Conjecture 3.4]. We prove Stafford’s conjecture:

AI comment INTRO-01. Proposed change — author acceptance pending. The quotient A_n/dA_n is always cyclic, generated by the class of 1. The nontrivial implication concerns general *finitely generated* torsion modules. Bellamy states the ordinary cyclicity question over $\mathbb{C}$; Stafford’s stable-cyclicity formulation is stronger. The red replacement above restores these distinctions.

Theorem 1.1. *For every characteristic-zero field k , every $n \geq 1$, and every nonzero $d \in A_n(k)$, there exist $F, R, S \in A_n(k)$ such that*

$$1 = dR + FdS.$$

Moreover, one may choose $F = \ell^{\deg_B d}$ for a linear Weyl coordinate ℓ , obtained by an invertible linear symplectic change of generators. Explicitly, $\ell = \Phi_g^{-1}(x_1)$ for a linear ω -preserving automorphism g as in Section 4.

Formal comparison: Lean also covers $n = 0$, where $A_0(k) = k$

The machine-checked proof of this result is registered in Palomar [5, 6]. Its Global Stafford extension [3, 4] proves the same-divisor identity $1 = dR + FdS$ in $D_k(A)$ for every smooth integral finitely generated characteristic-zero k -algebra A and every nonzero finite-order k -linear differential operator d . Polynomial A recovers the Weyl-algebra identity; the global theorem asserts existence of F , without the fixed linear-coordinate refinement in Theorem 1.1.

Proof sketch: Give each standard generator Bernstein degree one and let $N = \deg_B d$. A linear symplectic change of variables makes d monic of degree N in a momentum p , whose conjugate coordinate we denote by x . It is enough to prove that the cyclic right module

$$(1.1) \quad Q = A_n/(dA_n + x^N dA_n).$$

vanishes. An Euler identity makes x surjective on Q , while monicity makes $H = \{x = 0\}$ noncharacteristic and places $\text{Char}(Q)$ in a hypersurface $V(P)$ with $P(dx) \neq 0$. A filtered-page length argument shows that $\text{Char}(Q)$ has no point over H . Gabber’s theorem and the conormal argument of Sections 4 and 5 then force $\text{Char}(Q)$ to be empty. The same argument gives the more precise identity $A_n = dA_n + x^N dA_n$ in the chosen symplectic coordinates.

Constructive predecessors concern Stafford’s established generator theorems. Hillebrand–Schmale and Leykin give effective two-generator methods, and Leykin also constructs cyclic generators of holonomic modules [18, 21]. Quadrat–Robertz develop module-decomposition and basis algorithms [27, 28];

Caro–Levcovitz and Caro–Tuesta–Levcovitz study differential-operator rings with series coefficients [9, 10]. The constrained identity for arbitrary nonzero d requires the additional quotient-vanishing argument developed here.

The surrounding generator theory includes Stafford's stable-range and noncommutative Forster–Swan results [33, 32]. The generator article has a corrigendum and related exposition [35, 34]. Stafford also constructed stably free Weyl-algebra ideals [36]. OreModules supplies computational methods for modules over Ore algebras [11]. These provide context; the Euler residue, filtered-length contradiction, and asymptotic conormal construction below are project arguments.

2. ~~AI STATEMENT~~ [CHRIS] FORMAL VERIFICATION

~~Every statement of this paper, including Theorems 8.2 and 9.2 at the generality stated, is formalized and verified in Lean 4 against Mathlib, with no axiom beyond Lean's three standard ones.~~ The main identity, its fixed-source strengthening and the documented cyclicity and noncharacteristic consequences are verified in Lean 4 against Mathlib using only the three standard axioms. The formal proof also supplies the characteristic-support and geometric results used by its own argument. Statement scope and proof routes must be compared separately: machine verification of a terminal theorem does not certify every intermediate step of the printed proof. The mathematical differences are noted below; full formal comparisons and declaration links are in the technical supplement. The headline theorem is registered in Palomar [5]; the current Lean sources and auxiliary results are archived [6]. The accompanying proof-map supplement names the Lean declaration for each step.

AI comment WF-01. Proposed change — author acceptance pending. Please review each comparison at the stated generality. The Palomar record checks the registered main theorem; later results have separate archived sources. The formal geometric route uses Laurent series, while the printed tangent-limit argument is auxiliary. The short comparison notes identify these differences without claiming that every numbered statement or every proof passage has been verified verbatim.

The development uses Lean 4 [14], Mathlib [37] and AlgebraicAnalysis v0.3.0 [1]; tooling, chronology and statement comparison are described in Appendix A.

Online resources. Online Resource 1 is the interactive paper–Lean correspondence review, including the exact challenge definitions and archived proof endpoints; Online Resource 2 is the readable mathematical account of the selected formal proofs. Both are available from the [review companion](#), with a [frozen supplementary package](#) for journal review. The reusable HTML generator is maintained separately under the Apache 2.0 license; its software archive and the manuscript supplement have distinct version records. Proposed corrections and unresolved review actions remain explicitly marked in this draft.

3. NOTATION AND DEFINITIONS

Throughout, k is a field of characteristic zero and for the n -th Weyl algebra over k , we write

$$A_n(k) = k\langle x_1, \dots, x_n, \partial_1, \dots, \partial_n \rangle, \quad [\partial_i, x_j] = \delta_{ij}, \quad [x_i, x_j] = [\partial_i, \partial_j] = 0.$$

All ideals are *right* ideals and all modules are *right* modules unless the contrary is said explicitly.

Convention 3.1. We single out the first Weyl pair (x_1, ∂_1) and write

$$p := \partial_1, \quad C := A_{n-1}(k)[x_1], \quad A_n(k) = C[p; \partial_1],$$

so that x_1 is central in C , $[p, x_1] = 1$, and the transverse algebra $A_{n-1}(k)$ commutes with x_1 and p . Every element of $A_n(k)$ can uniquely be written as $\sum_j p^j c_j$ with $c_j \in C$. *This is the derivation Ore presentation; Ore[25] supplies its historical algebraic framework.*

Formal comparison: coefficients on the left, $\sum_j c_j p^j$; both normal forms are unique

Convention 3.2. *We define $H := \{x_1 = 0\} = \{(0, x_2, \dots, x_n)\} \subseteq \mathbb{A}^n$ as a distinguished hyperplane; its conormal covector is dx_1 , i.e. $\xi = (1, 0, \dots, 0)$. We write $\pi: T^*\mathbb{A}^n \rightarrow \mathbb{A}^n$ for the projection and identify ξ_i with dx_i , so that a covector is $\sum_i \xi_i dx_i$. The hyperplane $H = \{x_1 = 0\}$, the projection π and the identification $\xi_i \leftrightarrow dx_i$ are fixed in Section 3.1 below.*

AI comment DUP-01. Proposed change — author acceptance pending.
 Convention 3.2 repeated the definitions of H , π and ξ_i given again in Section 3.1; the red text keeps one copy. Alternatively delete the whole convention environment.

3.1. Cotangent space and symplectic structure. Let $X = \mathbb{A}_k^n$ be affine n -space over a field k of characteristic zero, with coordinates $x_1, \dots, x_n$. Its cotangent bundle T^*X is the space of pairs (y, ξ) consisting of a point $y \in X$ together with a covector ξ at y ; since X is affine space, this cotangent space is the same vector space $(k^n)^*$ at every point, and we may identify

$$T^*X \cong X \times (k^n)^*.$$

We write $\xi_1, \dots, \xi_n$ for the coordinates on $(k^n)^*$ dual to $x_1, \dots, x_n$, so that a covector is $\sum_i \xi_i dx_i$, and identify $\text{gr}^B A_n(k) = k[x, \xi]$ with the coordinate ring of T^*X . We write $\pi: T^*X \rightarrow X$ for the projection forgetting the covector, and for $y \in X$ write $T_y^*X = \pi^{-1}(y) \cong (k^n)^*$ for the fibre of T^*X over y .

We single out the first coordinate x_1 and define the distinguished hyperplane

$$H := \{x_1 = 0\} = \{(0, x_2, \dots, x_n)\} \subseteq X;$$

its conormal covector is dx_1 , i.e. $\xi = (1, 0, \dots, 0)$.

Definition 3.3 (Filtration [Def. 2.6 and 2.9 [29]]). *A filtration $F_\bullet = A_n(k)F_\bullet$ on $A_n(k)$ is a sequence of linear subspaces*

$$\{0\} = F_{-1} \subseteq F_0 \subseteq F_1 \subseteq \dots \subseteq A_n(k),$$

such that $F_i \cdot F_j \subseteq F_{i+j}$ and $A_n(k) = \bigcup F_j$. The associated graded algebra, also called symbol ring, is defined as

$$\mathrm{gr}^F A_n(k) = \bigoplus_{j=0}^{\infty} F_j / F_{j-1}.$$

Let $a \neq 0 \in A_n(k)$ and let j be the smallest integer, such that $a \in F_j$, called a filtration degree $\deg_F a$. ~~The~~ ~~the~~ ~~The~~ class of a in F_j / F_{j-1} is called its symbol $\sigma_j(a)$.

In the proof, we use two filtrations of $A_n(k)$.

- The Bernstein filtration $B_\bullet$ assigns degree 1 to every x_i and every ∂_i . The Bernstein degree of d is written $\deg_B d$.
- The order filtration $F_\bullet$ assigns degree 0 to every x_i and degree 1 to every ∂_i ; F_m is spanned by the monomials $x^\alpha \partial^\beta$ with $|\beta| \leq m$.

Formal comparison: as spans of ordered PBW words of bounded weight

For both the order filtration and the Bernstein filtration, the associated graded algebra $\mathrm{gr} A_n(k)$ is simply the polynomial ring $k[x, \xi] = k[x_1, \dots, x_n, \xi_1, \dots, \xi_n]$ in $2n$ variables with ξ_i the symbol of ∂_i . In particular, $\mathrm{gr} A_n(k)$ is commutative [29, Prop. 2.10]. Good filtrations and characteristic varieties are treated systematically in Björk and Hotta–Takeuchi–Tanisaki[8, 19].

Definition 3.4. A filtration of a right A_n -module M compatible with $F_\bullet$ is an exhaustive increasing chain of k -subspaces $M_\bullet$ with $M_i F_j \subseteq M_{i+j}$ and $M_i = 0$ for $i < 0$. It is good if the associated graded module $\mathrm{gr} M = \bigoplus_i M_i / M_{i-1}$ is finitely generated over $\mathrm{gr}^F A_n$.

As an example, the filtration $M_i := qF_i$ of a cyclic module over A_n induced by its generator q is good, since $\mathrm{gr} Q$ is generated by the element $\bar{q}$ denoting the class of q in Q_0 / Q_{-1} .

For a finitely generated right A_n -module M with a good order filtration $M_\bullet$, the graded module $\mathrm{gr} M$ is a finitely generated $k[x, \xi]$ -module.

Definition 3.5. The characteristic variety (also called symbol variety) of a finitely generated A_n -module M with a good order filtration $M_\bullet$ is defined by

$$\mathrm{Char}(M) := V(\mathrm{Ann}_{k[x, \xi]} \mathrm{gr} M) \subseteq T^* \mathbb{A}^n = \mathrm{Spec} k[x, \xi],$$

$\mathrm{Ann}_{k[x, \xi]} \mathrm{gr} M = \{f \in k[x, \xi] : f \cdot m = 0 \forall m \in \mathrm{gr} M\}$ are the annihilators of $\mathrm{gr} M$ over $k[x, \xi]$, and V denotes their vanishing locus, i.e. the set of points (x, ξ) for which all annihilator polynomials vanish. This set of points lies in the cotangent bundle $T^* \mathbb{A}^n$ of the affine n -space $\mathbb{A}^n$.

~~$\mathrm{Char}(M)$ is independent of the choice of the good filtration.~~ We only use $M = A_n / I$ with the filtration $M_i = qF_i$ induced by the class q of 1; then $\mathrm{Ann} \mathrm{gr} M$ is the order initial ideal of I , spanned by the order symbols of the elements of I . Points are prime ideals; for a non-closed field this is the correct reading of “the set of points”.

Formal comparison: for $M = A_n / I$

Definition 3.6 (Symplectic form and Poisson bracket). *A bilinear form ω on a k -vector space is symplectic if it is antisymmetric and nondegenerate.*

The standard symplectic form on the vector space k^{2n} , with basis $x_1, \dots, x_n, \partial_1, \dots, \partial_n$, is determined by the relations

$$\omega(x_i, \partial_j) = \delta_{ij}, \quad \omega(x_i, x_j) = \omega(\partial_i, \partial_j) = 0,$$

*analogously to the Weyl relations. Identifying $T^*X \cong X \times (\mathbb{C}^n)^*$ via the coordinates $x_1, \dots, x_n$ and their duals $\xi_1, \dots, \xi_n$, on T^*X this ω takes the familiar form*

$$\omega = \sum_{i=1}^n dx_i \wedge d\xi_i.$$

*The associated Poisson bracket on $k[x, \xi]$, the coordinate ring of T^*X , is*

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial \xi_i} \frac{\partial g}{\partial x_i} - \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial \xi_i} \right).$$

Formal comparison: Lean's bracket is the negative of this one; involutivity is unaffected

Definition 3.7 (involutive). *A reduced closed subset of $T^*\mathbb{A}^n$ is involutive if its vanishing ideal is closed under the Poisson bracket, $\{I(W), I(W)\} \subseteq I(W)$. An ideal J is involutive if $\{J, J\} \subseteq J$. This is weaker than being a Poisson ideal ($\{J, k[x, \xi]\} \subseteq J$): the ideal $(\xi_1, \dots, \xi_n)$ is involutive but not Poisson.*

AI comment ALG-01. Proposed change — author acceptance pending. Gabber applies to the radical characteristic ideal: put $J = \text{Ann}_{k[x, \xi]} \text{gr } M$ and $I = \sqrt{J}$; then $\{I, I\} \subseteq I$. The set $\text{Char}(M) = V(J)$ is not an ideal. Use $I = \sqrt{J}$ in the componentwise lemma and its proof below. Minimal primes of J and $\sqrt{J}$ coincide, so the localization argument then has the intended hypotheses.

3.2. Gabber's theorem.

Theorem 3.8 (Gabber [15]; algebraic form in SinghKumar[30]; see also [19, §2.4Theorem 2.3.1 and Appendix D], Björk). *Let I be any right ideal of $A_n(k)$, with k a field of characteristic zero, and give $M = A_n(k)/I$ its order quotient filtration. Then, ~~the characteristic variety $\text{Char}(M)$ is an involutive ideal, i.e. $\{\text{Char}(M), \text{Char}(M)\} \subseteq \text{Char}(M)$.~~ the radical $\sqrt{J}$ of $J = \text{Ann}_{k[x, \xi]} \text{gr } M$ is involutive, $\{\sqrt{J}, \sqrt{J}\} \subseteq \sqrt{J}$; equivalently, $\text{Char}(M) = V(J)$ is involutive in the sense of Definition 3.7.*

Formal comparison: proved in Lean, not cited; exactly this scope, over every field of characteristic zero

AI comment LIT-01. Information — no edit requested. Statement check: Gabber [15] proves that $\sqrt{\text{Ann gr } M}$ is closed under the Poisson bracket for every finitely generated module M with a good filtration over a filtered ring whose associated graded ring is commutative, Noetherian and contains $\mathbb{Q}$; Singh–Kumar give the algebraic proof. The red statement above is the special case $M = A_n(k)/I$ with the quotient filtration, which is all the paper uses and exactly what the formalization proves (following the Singh–Kumar argument: Rees ring modulo T^2 , localization at a minimal prime, Artinian truncation, trace identity).

Lemma 3.9 (Componentwise involutivity). *Let M be a finitely generated $A_n(k)$ -module with a good filtration, and let $\text{Char}(M)$ be its characteristic variety, so that by Gabber's theorem $\{\text{Char}(M), \text{Char}(M)\} \subseteq \text{Char}(M)$. If $\mathfrak{p}$ is a minimal prime of $\text{Ann}_{k[x,\xi]} \text{gr } M$ (equivalently, a minimal prime over $\text{Char}(M)$), let J be an ideal of $k[x,\xi]$ whose radical is involutive (by Theorem 3.8, e.g. $J = \text{Ann gr } M$ for $M = A_n(k)/I$). If $\mathfrak{p}$ is a minimal prime of J , then $\mathfrak{p}$ is itself involutive:*

$$\{\mathfrak{p}, \mathfrak{p}\} \subseteq \mathfrak{p}.$$

Proof. Write $R = k[x,\xi]$ and $I = \sqrt{\text{Char}(M)J}$. Localize at $\mathfrak{p}$. Since $\mathfrak{p}$ is minimal over I , the only prime of $R_{\mathfrak{p}}$ containing $I_{\mathfrak{p}}$ is $\mathfrak{p}_{\mathfrak{p}}$ itself; as I is already radical, so is its localization, giving

$$I_{\mathfrak{p}} = \sqrt{I_{\mathfrak{p}}} = \mathfrak{p}_{\mathfrak{p}}.$$

The Poisson bracket extends to $R_{\mathfrak{p}}$: by the Leibniz rule, $\{a/s, b/t\}$ is determined by $\{a, b\}$ and the brackets of a, b with s, t , giving a well-defined bracket on $R_{\mathfrak{p}}$ agreeing with the one on R under $f \mapsto f/1$; in particular $\{f/1, g/1\} = \{f, g\}/1$ for $f, g \in R$, and the involutivity of I localizes to $\{I_{\mathfrak{p}}, I_{\mathfrak{p}}\} \subseteq I_{\mathfrak{p}}$.

Now take $f, g \in \mathfrak{p}$. Then $f/1, g/1 \in \mathfrak{p}_{\mathfrak{p}} = I_{\mathfrak{p}}$, so

$$\{f, g\}/1 = \{f/1, g/1\} \in I_{\mathfrak{p}} = \mathfrak{p}_{\mathfrak{p}}.$$

Thus there is $t \notin \mathfrak{p}$ with $t\{f, g\} \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime and $t \notin \mathfrak{p}$, this forces $\{f, g\} \in \mathfrak{p}$. $\square$

Formal comparison: any ideal with involutive radical; proof without localization

AI comment COMP-01. Information — proof comparison; no edit requested. The formal argument starts with an involutive radical and proves that each minimal prime is involutive directly, without localization. Thus the hypothesis is radical involutivity, not that the whole ideal is a Poisson ideal. **Mathematical proof** (COMP-01, p. 1).

4. MAPPING TO A MONIC NORMAL FORM

Given a fixed nonzero $d \in A_n(k)$ of Bernstein degree N , our first goal is to find a map to replace d by a normal form $\tilde{d}$ that is monic in p , i.e. of the

form $\tilde{d} = p^N + \sum_{j < N} p^j a_j$ with coefficients a_j in C .

If $N = 0$, then $d \in B_0 = k$ is a nonzero scalar, hence a unit, and $1 = dR + FdS$ holds trivially with $R = d^{-1}$, $F = 0$, $S = 0$. We therefore assume $N \geq 1$.

AI comment ALG-07. Proposed change — author acceptance pending.
In the scalar case choose $F = 1$, $R = d^{-1}$, $S = 0$ to satisfy the stronger assertion $F = \ell^0 = 1$ too.

We want to construct a well-behaved automorphism of $\text{gr}^B A_n(k) = k[x, \xi]$ from simpler linear algebra. We proceed in three stages: define an automorphism of the vector space $\mathbb{V} = k^{2n}$, extend it to an automorphism of $A_n(k)$ itself, and observe the induced action this extension leaves on the associated graded ring.

We can identify $\mathbb{V} = k^{2n}$ with the Bernstein-degree-one part B_1/B_0 of $A_n(k)$, i.e. linear combinations of $x_1, \dots, x_n, \partial_1, \dots, \partial_n$; the coordinates of a vector are then the coefficients of these generators. By this identification, $\mathbb{V}$ carries the standard symplectic form ω (Def. 3.6). Let g be a k -linear automorphism of $\mathbb{V}$ that preserves the symplectic form, that is $\omega(gu, gv) = \omega(u, v)$ for all $u, v \in \mathbb{V}$.

Every such g extends uniquely to a k -algebra automorphism Φ_g of $A_n(k)$: substituting each generator $u \in \mathbb{V}$ by its image $g(u)$ keeps the Weyl relations $[u, v] = \omega(u, v)[v, u] = \omega(u, v)$ (recall $[\partial_i, x_i] = 1 = \omega(x_i, \partial_i)$) intact exactly when g preserves ω , and this substitution on generators determines Φ_g on all of $A_n(k)$.

Finally, since Φ_g sends each generator to another degree-one element, it preserves the Bernstein filtration, and therefore induces a well-defined automorphism of the associated graded ring $\text{gr}^B A_n(k) = k[x, \xi]$. ~~Concretely, this induced map sends a symbol (a polynomial function on $\mathbb{V}$) s to $s \circ g^{-1}$: functions transform by precomposition with the inverse, since Φ_g moves the underlying points of $\mathbb{V}$ forward by g .~~ Concretely, write the generators as $z_1, \dots, z_{2n}$ and let G be the matrix of g , $g(z_i) = \sum_j G_{ji} z_j$. A symbol is a polynomial $s(z)$, and Φ_g substitutes $z_i \mapsto \sum_j G_{ji} z_j$. Reading s as a function of coefficient vectors $c \in k^{2n}$, the symbol of $\Phi_g(d)$ is therefore $c \mapsto s(G^T c)$, i.e. $s \circ g^T$ (not $s \circ g^{-1}$). For homogeneous s of degree N , the value at the basis vector e_p is the coefficient of z_p^N .

AI comment ALG-08. Proposed change — author acceptance pending.
The printed law $s \mapsto s \circ g^{-1}$ is the point convention $\Phi(z) = z \circ g^{-1}$, not the substitution $\Phi_g(u) = g(u)$ defined above. Counterexample to the printed proof of Proposition 4.2: $n = 1$, $d = x$, $v = x + \partial$ (so $s(v) = 1$), $g = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$, which preserves ω and maps v to $e_p = \partial$; yet $\Phi_g(x) = x$ has no p -term. The red replacement and the corrected proof below use $s \circ g^T$, as the formalization does.

Lemma 4.1. *Let $(\mathbb{V}, \omega)$ be a symplectic k -space of dimension $2n$ and $0 \neq v \in \mathbb{V}$. Let $\mathbb{V} = k^{2n}$ carry the standard symplectic form fixed above, and let $0 \neq v \in \mathbb{V}$.*

~~Then v is the first vector of a symplectic basis of $\mathbb{V}$; equivalently, there exists a k -linear g preserving ω with $g(v) = e$ for any prescribed nonzero e .~~ For any prescribed nonzero e , there exists a k -linear automorphism g preserving ω with $g(v) = e$.

Proof. Since ω is nondegenerate and $v \neq 0$, the functional $\omega(v, -)$ is not identically zero, so there is $w' \in \mathbb{V}$ with $\omega(v, w') = c \neq 0$; setting $w := w'/c$ gives $\omega(v, w) = 1$.

Let $U = \text{span}(v, w)$. For $0 \neq av + bw \in U$: if $a \neq 0$, $\omega(av + bw, w) = a \neq 0$; if $b \neq 0$, $\omega(av + bw, v) = -b \neq 0$. Either way $av + bw$ pairs nontrivially with something in U , so $\omega|_U$ is nondegenerate.

Since $\omega|_U$ is nondegenerate, $U \cap U^\perp = 0$, and as $\dim U + \dim U^\perp = \dim \mathbb{V}$ for a nondegenerate form, this gives $\mathbb{V} = U \oplus U^\perp$. If $u \in U^\perp$ pairs to zero with all of $U^\perp$, it also pairs to zero with U (as $u \in U^\perp$), hence with all of $\mathbb{V}$, forcing $u = 0$; so $\omega|_{U^\perp}$ is nondegenerate too.

By induction on n (trivial for $n = 1$, where $U = \mathbb{V}$), a symplectic basis $e_2, \dots, f_n$ of $U^\perp$ exists; setting $e_1 := v$, $f_1 := w$ gives a symplectic basis of $\mathbb{V} = U \oplus U^\perp$, since cross-pairings between U and $U^\perp$ vanish by definition of $U^\perp$.

For the second formulation: given nonzero $v, e \in \mathbb{V}$, complete each to a symplectic basis as above. The linear map g sending one basis to the other, vector by vector, preserves all pairwise values of ω on basis vectors, hence by bilinearity preserves ω on all of $\mathbb{V}$; and $g(v) = e$. $\square$

Formal comparison: second formulation only; the formal proof composes at most two symplectic transvections

AI comment COMP-02. Proposed change — author acceptance pending.

The red statement specializes to the standard symplectic space used by the paper and retains the transitivity formulation proved by the formal endpoint. The arbitrary-space formulation would additionally require a checked coordinate-identification bridge. The stronger basis sentence remains in blue as an unselected alternative; restoring it requires its own formal correspondence. The formal proof establishes transitivity on nonzero vectors by at most two symplectic transvections. It does not prove the separate symplectic-basis formulation. Reversing the sign of the alternating form leaves its isometry group unchanged. [Mathematical proof \(COMP-02, p. 1\)](#).

Proposition 4.2. *Let $0 \neq d \in A_n(k)$ with $N := \deg_B d \geq 1$. There exists a k -linear ω -preserving automorphism g and $c \in k^\times$ such that*

$$(4.1) \quad \tilde{d} := c^{-1} \Phi_g(d) = p^N + \sum_{j < N} p^j a_j, \quad a_j \in C, \quad \deg_B a_j \leq N - j,$$

in the notation of Convention 3.1.

Proof. Let $s \in k[x, \xi]$ be the Bernstein principal symbol of d , a nonzero homogeneous polynomial of degree N on $\mathbb{V}$. A characteristic-zero field is infinite, so every nonzero polynomial has a nonvanishing value: choose

$v \in \mathbb{V}$ with $s(v) \neq 0$. By Lemma 4.1 we can map v to the basis vector e_1 in p -direction via an automorphism g . The Bernstein symbol of $\Phi_g(d)$ is $s \circ g^{-1}$, whose value at e_1 is $s(v) \neq 0$. By Lemma 4.1 there is an ω -preserving h with $h(e_1) = v$, where e_1 is the basis vector in the p -direction. Let g be the map with matrix $H^\top$; the transpose of a symplectic matrix is symplectic, so g preserves ω . The Bernstein symbol of $\Phi_g(d)$ is $s \circ g^\top = s \circ h$, whose value at e_1 is $s(v) \neq 0$. Put $c = s(v)$; since c is a nonzero scalar, it is a central unit of $A_n(k)$, and $\tilde{d} = c^{-1}\Phi_g(d)$ has its Bernstein symbol taking the value 1 at e_1 . Since e_1 has coordinate 1 in the p -direction and 0 elsewhere, that value is precisely the coefficient of the monomial p^N in $\tilde{d}$. Collecting the remaining terms by their p -degree and writing the coefficients on the right, using the unique normal form $A_n(k) = \bigoplus_j p^j C$, gives $\tilde{d} = p^N + \sum_{j < N} p^j a_j$ with $a_j \in C$. The Bernstein bound $\deg_B a_j \leq N - j$ holds because $\deg_B \tilde{d} = N$ and Φ_g preserves $\deg_B$; no cancellation between different p -degrees occurs, because the normal form $\bigoplus_j p^j C$ is unique. $\square$

Formal comparison: monicity is stated as: $\tilde{d} \in B_N$ and the PBW coefficient of ∂_1^N is 1

AI comment COMP-03. Information — proof comparison; no edit requested. The formal normalization uses an explicit symplectic matrix and its inverse. Its symbol substitution is $s(MX)$; with the convention used here this is $s \circ g^\top$. Monicity means that the p^N PBW coefficient is 1 and the Bernstein degree is at most N , which gives the coefficient bounds above. [Mathematical proof \(COMP-03, p. 2\).](#)

All coefficients involved belong to a finitely generated subfield $k_0 \subset k$, namely the subfield generated by the finitely many coefficients of d in the new generators. If we find $R, S, F \in A_n(k_0)$ sufficing Theorem 1.1, the same choices extend to k as well, so proving the identity over k_0 is sufficient. We therefore replace k by k_0 , write $A_n := A_n(k_0)$, and fix an embedding $k_0 \hookrightarrow \mathbb{C}$. Sections 4–7 work over k itself; to keep the printed formulas we write $k_0 := k$ and $A_n := A_n(k_0)$ there. Sections 8 and 9 work over an algebraically closed field K of characteristic zero; the reader may take $K = \mathbb{C}$, and Section 10 uses $K = \bar{k}$. No embedding of k into $\mathbb{C}$ is needed.

AI comment FIELD-01. Proposed change — author acceptance pending. The formalization never reduces to a finitely generated subfield: it runs Sections 8–9 over $\bar{k}$ and descends the order initial ideal (Section 10). The red text adopts this route; the struck text is no longer needed.

Remark 4.3. Φ_g is a ring automorphism, so it preserves written order term by term. Consequently an identity $1 = \tilde{d}R + x_1^N \tilde{d}S$ transports back to

$$1 = d(c^{-1}\Phi_g^{-1}(R)) + F d(c^{-1}\Phi_g^{-1}(S)), \quad F = \Phi_g^{-1}(x_1)^N,$$

using that c^{-1} is a central scalar.

We now use the order filtration on $A_n(k_0)$, where coordinates have degree zero and derivatives have degree one, to characterize $P = \sigma_{\text{ord}}(\tilde{d})$, the order principal symbol of $\tilde{d}$. Recall that a symbol, being the class of an element in a single graded piece, is automatically *homogeneous* of the corresponding degree, i.e. contains only terms of that total degree.

Definition 4.4. *We say a polynomial $P = \sum_{i=0}^N c_i \xi_1^i \in k_0[\xi_1, \dots, \xi_n]$, written as a polynomial in ξ_1 with coefficients $c_i \in k_0[\xi_2, \dots, \xi_n]$, is monic of degree N in ξ_1 if $c_N = 1$.*

Lemma 4.5. *Let $\tilde{d} = p^N + \sum_{j < N} p^j a_j$ with $a_j \in C$, $\deg_B a_j \leq N - j$, as in (4.1), and set $N = \deg_B \tilde{d}$. Then $\tilde{d}$ has order N , and its order principal symbol is a polynomial in ξ alone,*

$$(4.2) \quad \sigma_{\text{ord}}(\tilde{d}) = P(\xi), \quad P \in k_0[\xi_1, \dots, \xi_n], \quad \deg P = N.$$

P is monic of degree N in ξ_1 and in particular

$$(4.3) \quad P(1, 0, \dots, 0) = 1.$$

Proof. Since $\deg_B \tilde{d} = N$, every PBW monomial $x^\alpha \partial^\beta$ occurring in $\tilde{d}$ satisfies $|\alpha| + |\beta| \leq N$. A monomial contributing to the order- N part has $|\beta| = N$, hence $|\alpha| = 0$: it is a pure ∂^β with scalar coefficient. Thus $\tilde{d}$ has order N , and its order- N part is a k_0 -linear combination of ∂^β with $|\beta| = N$, giving $\sigma_{\text{ord}}(\tilde{d}) = P(\xi)$, homogeneous of degree N with coefficients in k_0 .

The monomial $p^N = \partial_1^N$ occurs in $\tilde{d}$ with coefficient 1; every other order- N monomial comes from some $p^j a_j$ with $j < N$ and $a_j \in C$ (hence involving no further factor of $p = \partial_1$), so it carries a positive power of some transverse ∂_i , $i > 1$. Such a monomial contributes only to coefficients c_i with $i < N$, never to c_N ; hence c_N receives no contribution besides the coefficient of p^N itself, giving $c_N = 1$ identically, i.e. P is monic of degree N in ξ_1 . Evaluating at $(1, 0, \dots, 0)$ then gives $P(1, 0, \dots, 0) = 1$. $\square$

5. x_1 AS A SURJECTIVE MAP

We define $E = x_1 p$ and

$$(5.1) \quad I = \tilde{d}A_n + x_1^N \tilde{d}A_n, \quad Q = A_n/I.$$

The goal of this subsection will be to show that $Q = Qx_1$ (Proposition 5.3).

The adjoint action of E defines a grading of A_n : x_1 has Euler degree 1, p has Euler degree -1 , and the transverse Weyl algebra has Euler degree zero. Let $A_{\geq 0}$ denote the nonnegative part of A_n in this grading. In PBW form (with x_i on the left of ∂_i), $A_{\geq 0}$ is generated by the transverse algebra, E , and x_1 . Indeed A_n has the basis $b x_1^a p^c$ with b transverse, of Euler degree $a - c$; if $a \geq c$ then $x_1^a p^c = x_1^{a-c} (x_1^c p^c)$ and $x_1^c p^c$ is a polynomial in E by Lemma 5.1. In particular every element of Euler degree ≥ 1 lies in $A_{\geq 0} x_1$,

using $Ex_1 = x_1(E+1)$ to move x_1 to the right. By the commutation relations of the Weyl algebra, it holds that

$$(5.2) \quad \begin{aligned} Ex_1 &= x_1(E+1), & x_1E &= (E-1)x_1, \\ Ep &= p(E-1), & pE &= (E+1)p \end{aligned}$$

Additionally, x_1 commutes with the transverse algebra, so, together with $x_1E = (E-1)x_1$, x_1 is normal in $A^{\geq 0}$.

Formal comparison: Lean uses the subring generated by the transverse algebra, x_1 and x_1p , and one-sided normality $x_1r = r'x_1$, which is all that is used

AI comment ALG-02. Proposed change — author acceptance pending.
The displayed products below have reversed shifts. For $E = x_1p$ and $[p, x_1] = 1$, the correct identities are $x_1^m p^m = \prod_{i=0}^{m-1} (E-i)$ and $p^m x_1^m = \prod_{i=1}^m (E+i)$. Already $m = 1$ gives $x_1p = E$, $px_1 = E+1$. Adjust the induction accordingly; the next residue proof already uses these correct products.

Lemma 5.1. *For every $m \geq 0$,*

$$(5.3) \quad x_1^m p^m = \prod_{i=1}^m (E-i) \prod_{i=0}^{m-1} (E-i), \quad p^m x_1^m = \prod_{i=0}^{m-1} (E+i) \prod_{i=1}^m (E+i),$$

the factors in each product commuting with one another.

Proof. Since both products in (5.3) are polynomials in the single element E , their factors commute. We induct on m ; both identities are trivial for $m = 0$. Assuming the first identity for m , $x_1^{m+1} p^{m+1} = x_1 \left(\prod_{i=1}^m (E-i) \right) p = \left(\prod_{i=1}^m (E-1-i) \right) x_1 p = \left(\prod_{i=1}^m (E-i) \right) E = \prod_{i=0}^m (E-i)$,
 $= \left(\prod_{i=1}^m (E-1-i) \right) x_1 p = \left(\prod_{i=2}^{m+1} (E-i) \right) (E-1) = \prod_{i=1}^{m+1} (E-i)$,

$$x_1^{m+1} p^{m+1} = x_1 \left(\prod_{i=0}^{m-1} (E-i) \right) p = \left(\prod_{i=0}^{m-1} (E-1-i) \right) x_1 p = \left(\prod_{i=1}^m (E-i) \right) E = \prod_{i=0}^m (E-i),$$

using $x_1E = (E-1)x_1$ repeatedly and the definition of E . The second identity follows identically from $Ep = p(E-1)$:

$$p^{m+1} x_1^{m+1} = p \left(\prod_{i=1}^m (E+i) \right) x_1 = \left(\prod_{i=1}^m (E+1+i) \right) p x_1 = \left(\prod_{i=2}^{m+1} (E+i) \right) (E+1).$$

□

Lemma 5.2. *There exists an element $U \in A_{\geq 0}$ such that*

$$(5.4) \quad 1 + Ux_1 \in I.$$

Proof. From Lemma 5.1 we define

$$p^N x_1^N = \prod_{r=1}^N (E+r) =: R_+(E), \quad x_1^N p^N = \prod_{r=0}^{N-1} (E-r) =: R_-(E).$$

The roots of R_+ and R_- are disjoint (the sets $\{0, \dots, 1-N\}$ and $\{1, \dots, N\}$ of integers, embedded in the

prime field $\mathbb{Q} \subset k_0$), so $\gcd(R_+, R_-) = 1$ in $\mathbb{Q}[E]$. By the extended Euclidean algorithm, there exist $u, v \in \mathbb{Q}[E] \subset k_0[E]$ with

$$R_+(E)u(E) + R_-(E)v(E) = p^N x_1^N u(E) + x_1^N p^N v(E) = 1.$$

We write $\tilde{d} = p^N + \Sigma$ with $\Sigma = \sum_{j < N} p^j a_j$. Then

$$(\tilde{d} x_1^N)u(E) + (x_1^N \tilde{d})v(E) = 1 + w \in I,$$

where

$$w = \Sigma x_1^N u(E) + x_1^N \Sigma v(E).$$

We can write each a_j in Σ in powers of x_1 , with coefficients in the transverse Weyl algebra. A term $b x_1^r p^j$ coming from ~~$a_j p^j p^j a_j$~~ (Euler degree does not depend on the order of the factors) has $r \geq 0$ and $j < N$, so ~~and~~ multiplication by x_1^N gives that term Euler degree at least $N - j > 0$. u and v only depend on E , so they have Euler degree 0. Therefore every term of w has positive Euler degree, and thus $w \in A_{\geq 0} x_1$ (by the remark after (5.2)) - or equivalently, there exists a $U \in A^{\geq 0}$, such that $w = U x_1$. $\square$

Formal comparison: the ring identity is Lean's; the factorization uses left coefficients $c x^a p^j$

AI comment COMP-04. Information — proof comparison; no edit requested. The formal argument replaces the Euler grading by the subring generated by the transverse algebra, x_1 and $E = x_1 p$. It needs only one-sided normality $x_1 r = r x_1$ and left-coefficient normal forms. The corrected Euler product identities give a polynomial annihilator with nonzero constant term; no two-sided normality is asserted. **Mathematical proof (COMP-04, p. 2).**

Proposition 5.3. *Right multiplication by x_1 on Q is surjective.*

Proof. Let q be the class of 1 in Q . By Equation (5.4), we have $q = -q U x_1$, so $q \in (q A_{\geq 0}) x_1$. We use the normality of x_1 to show

$$(5.5) \quad q A_{\geq 0} = (q A_{\geq 0}) x_1.$$

The inclusion $\supseteq$ holds because $x_1 \in A^{\geq 0}$ and $A^{\geq 0}$ is a ring. For $\subseteq$, let $r \in A^{\geq 0}$ and choose r' with $x_1 r = r' x_1$ (possible since x_1 is normal) then $q r = -(q U) x_1 r = -(q U r') x_1 \in (q A^{\geq 0}) x_1$ since $U r' \in A^{\geq 0}$.

$q A_{\geq 0}$ is stable under right multiplication by p : If $m = m' x_1$, then

$$m p = m' (x_1 p) = m' E \in q A_{\geq 0}.$$

It is of course also stable under C , since ~~elements of C have Euler degree 0~~ $\theta C = A_{n-1}(k)[x_1] \subseteq A_{\geq 0}$. Since C, p generate A_n , we have $q A_{\geq 0} = q A_n = Q$. Equation (5.5) is therefore $Q = Q x_1$.

AI comment ALG-03. Proposed change — author acceptance pending. $C = A_{n-1}(k)[x_1]$ is not entirely of Euler degree zero: x_1 has degree 1. The needed stability still holds because both the transverse algebra and x_1 belong to $A_{\geq 0}$, so $C \subseteq A_{\geq 0}$. Replace the degree-zero explanation above.

$\square$

AI comment COMP-05. Information — proof comparison; no edit requested. The formal proof works element by element with the transverse–Euler subring. The residue identity gives $1 + Ux_1 \in I$ for a suitable U in that subring; one-sided normality and the monic relation then give surjectivity of right multiplication by x_1 . This avoids treating a quotient by a vector subspace as an algebra quotient. [Mathematical proof \(COMP-05, p. 3\)](#).

6. VANISHING COKERNEL

In this section and the next, C denotes the commutative symbol ring $k_0[x, \xi] = T[x_1, \xi_1]$ (not the Ore coefficient algebra of [Convention 3.1](#)), and [Corollary 7.1](#) of [Section 7](#), whose proof uses only [Lemma 4.5](#), is used freely.

AI comment NOTATION-01. Proposed change — author acceptance pending. C below denotes the commutative symbol ring $T[x_1, \xi_1]$, whereas [Convention 3.1](#) used C for the noncommutative coefficient algebra $A_{n-1}(k)[x_1]$. Please rename one or explicitly announce the change of notation.

Put $M = \text{gr } Q$ and

$$T = k_0[x_2, \dots, x_n, \xi_2, \dots, \xi_n], \quad C = T[x_1, \xi_1],$$

$$V = \ker(R_{x_1} : M \rightarrow M), \quad U = \text{coker}(R_{x_1} : M \rightarrow M) = M/x_1M.$$

With R_{x_1} being the right multiplication ~~with~~by x_1 . If $n = 1$, we define $T = k_0$. We want to prove that $U = 0$ by comparing two lengths over a localization of T .

Definition 6.1 ([\[13, 12\]](#)[\[38, Tags 02LY, 00J3, 00IV\]](#)). *The length ℓ_T of a ~~finite T -module~~ T -module of finite length M is defined as the number of steps in a composition series of M such that*

$$0 = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_{\ell_T} = M$$

where each successive quotient M_i/M_{i-1} is simple (i.e. has no proper nontrivial submodules). The length is well-defined, as it does not depend on the choice of the composition series. Lengths are additive for short exact sequences.

Lemma 6.2. *The modules U and V are finite over T , and $\text{Supp}_T V \subseteq \text{Supp}_T U$. If $\mathfrak{q}$ is a minimal prime in $\text{Supp}_T U$, then $U_{\mathfrak{q}}$ and $V_{\mathfrak{q}}$ have finite length over $T_{\mathfrak{q}}$.*

Proof. The good quotient filtration makes $M = \text{gr } Q$ finitely generated over $C = \text{gr } A_n = k_0[x, \xi] = T[x_1, \xi_1]$. Since P annihilates M ([Corollary 7.1](#)), this action factors through $C/(P)$, which, because P is monic of degree N in ξ_1 , is generated over $T[x_1]$ by $1, \xi_1, \dots, \xi_1^{N-1}$. Thus M is finite over $T[x_1]$. Because M is finite, and $T[x_1]$ is a Noetherian ring, the kernel V and cokernel U of multiplication in M by x_1 are also finite over $T[x_1]$. Since they are annihilated by x_1 , they are finite over T .

Suppose $\mathfrak{p}$ is a prime of T where $U_{\mathfrak{p}} = (M/x_1M)_{\mathfrak{p}} = (M_{\mathfrak{p}}/x_1M_{\mathfrak{p}}) = 0$. Consequently, $M_{\mathfrak{p}} = x_1M_{\mathfrak{p}}$, i.e. multiplication by x_1 is a surjective endomorphism of the finite module $M_{\mathfrak{p}}$ over the corresponding localization

of C . By [Vasconcelos's theorem \[22\]](#)[\[Theorem 2.4\]](#)[Vasconcelos' theorem \[22, Theorem 2.4\]](#), this endomorphism is also injective, which means $V_{\mathfrak{p}} = \ker(x_1 : M_{\mathfrak{p}} \rightarrow M_{\mathfrak{p}}) = 0$, i.e. $\mathfrak{p} \notin \text{Supp}_T(V)$, and therefore $\text{Supp}_T V \subseteq \text{Supp}_T U$.

After localization at a minimal support prime $\mathfrak{q}$ of U , both finite T -modules have support contained in the closed point. They therefore have finite length. $\square$

AI comment ALG-04. Max — correspondence check; no rewrite of this lemma requested. The monic quotient argument above now supplies the finite- T -module bridge for U, V . The earlier request for that argument is resolved. In the later page argument, check the T -action and T -linear maps: localize the associated graded pages, not Q . See COMP-06.

Lemma 6.3. *For every minimal prime $\mathfrak{q}$ in $\text{Supp}_T U$,*

$$\ell_{T_{\mathfrak{q}}}(U_{\mathfrak{q}}) \leq \ell_{T_{\mathfrak{q}}}(V_{\mathfrak{q}}).$$

Proof. To prove the inequality we set up a spectral sequence measuring how $\ker(R_{x_1})$ and $\text{coker}(R_{x_1})$ compare, order by order, to $V = \ker(x : M \rightarrow M)$ and $U = M/xM$.

Consider the filtered complex $Q \xrightarrow{R_{x_1}} Q$ (right multiplication by x_1 , between two copies of Q), with decreasing filtration $G^p = F_{-p}Q$, where F is the order filtration. Since x_1 has order 0, $R_{x_1}(G^p) \subseteq G^p$ for every p ; the induced map on G^p/G^{p+1} , summed over p , is exactly the action of x_1 on $M = \text{gr } Q$. This is the crudest approximation to $\ker(R_{x_1})$ and $\text{coker}(R_{x_1})$: it correctly identifies V, U as approximations, but it does not yet see elements that vanish to higher order than this naive count predicts. The spectral sequence refines this approximation in successive stages, called *pages*: for each $r \geq 0$, the source and target pages are

$$A_r = \bigoplus_{p \in \mathbb{Z}} \frac{G^p \cap R_{x_1}^{-1}(G^{p+r})}{G^{p+1} \cap R_{x_1}^{-1}(G^{p+r})}, \quad B_r = \bigoplus_{p \in \mathbb{Z}} \frac{G^p}{(G^p \cap R_{x_1}(G^{p-r+1})) + G^{p+1}},$$

where the degree- p summand of A_r isolates elements of G^p whose image under R_{x_1} drops r extra filtration levels (candidates for the kernel, accurate to order r), and the degree- p summand of B_r isolates the part of G^p not yet hit by R_{x_1} from $r-1$ levels shallower.

Applying R_{x_1} to a representative gives a well-defined map $d_r : A_r \rightarrow B_r$, shifting the summand index by r ; one checks directly from these definitions that

$$A_0 = B_0 = M, \quad d_0 = x_1, \quad A_{r+1} = \ker d_r, \quad B_{r+1} = \text{coker } d_r,$$

so that each page is obtained from the previous one by taking kernel and cokernel of d_r , exactly as in the construction of a spectral sequence from a filtered complex.

Now we fix a minimal prime $\mathfrak{q}$ in $\text{Supp}_T U$ and localize at $T \setminus \mathfrak{q}$. Since localization is exact, it commutes with kernels, cokernels, intersections, and preimages, and hence with every operation used to define A_r, B_r and d_r from

G^p and R_{x_1} . Consequently the localized objects $(A_r)_q, (B_r)_q$ are again the pages of the spectral sequence built the same way from the localized complex $Q_q \xrightarrow{R_{x_1}} Q_q$. By Lemma 6.2, $A_1 = V_q$ and $B_1 = U_q$ have finite length over T_q ; since each $A_{r+1} \subseteq A_r$ and B_{r+1} is a quotient of B_r , all later pages have finite length as well.

From here on, all lengths are taken over T_q , and we write A_r, B_r, U, V for the localized objects $(A_r)_q, (B_r)_q, U_q, V_q$.

The maps $B_1 \rightarrow B_r$ are surjective by construction. Let $K_r := \ker(B_1 \rightarrow B_r)$. Since $B_1 \rightarrow B_{r+1}$ factors through $B_1 \rightarrow B_r \rightarrow B_{r+1}$, we have $K_r \subseteq K_{r+1}$, so $(K_r)_{r \geq 1}$ is an increasing chain of submodules of B_1 . This chain exhausts B_1 : given $u \in B_1 = U$, choose a representative in Q_q . Since localization is exact, the surjection $Q = Q_{x_1}$ of Proposition 5.3 localizes to $Q_q = Q_q x_1$, so this representative has an x_1 -preimage in Q_q ; the same holds for that preimage, and so on. Since the order filtration on Q_q (induced from the order filtration on Q) is exhaustive, any fixed element lies in some finite filtration step, and this iterated-preimage process can be carried out only finitely many times before the resulting class becomes zero in the corresponding page. Concretely, this shows that u maps to 0 in B_r for some r depending on u , i.e. $u \in K_r$. As $u \in B_1$ was arbitrary, $\bigcup_{r \geq 1} K_r = B_1$. Since B_1 has finite length, it is a Noetherian T_q -module. By the ascending chain condition, the increasing chain $K_1 \subseteq K_2 \subseteq \dots$ stabilizes: there exists a finite R with $K_R = K_{R+i}$ for all $i \geq 0$. Combined with $\bigcup_r K_r = B_1$, stabilization forces $K_R = B_1$. Finally, $K_R = B_1$ means $B_1 \rightarrow B_R$ must be the zero map: by the first isomorphism theorem, $B_1/K_R \cong \text{im}(B_1 \rightarrow B_R)$, and $K_R = B_1$ gives $B_1/K_R = 0$, so $\text{im}(B_1 \rightarrow B_R) = 0$. But $B_1 \rightarrow B_R$ is also surjective, so $B_R = 0$.

At each stage $r \geq 1$ the differential d_r gives an exact sequence

$$0 \rightarrow A_{r+1} \rightarrow A_r \xrightarrow{d_r} B_r \rightarrow B_{r+1} \rightarrow 0,$$

which splits at $\text{im}(d_r)$ into two short exact sequences. Additivity of length gives $\ell(A_r) = \ell(A_{r+1}) + \ell(\text{im } d_r)$ and $\ell(B_r) = \ell(\text{im } d_r) + \ell(B_{r+1})$; eliminating $\ell(\text{im } d_r)$ yields, for every $r \geq 1$,

$$(6.1) \quad \ell(B_r) + \ell(A_{r+1}) = \ell(A_r) + \ell(B_{r+1}).$$

Summing this identity over $r = 1, \dots, R-1$ and reindexing the sums $\sum_{r=1}^{R-1} \ell(A_{r+1}) = \sum_{r=2}^R \ell(A_r)$, $\sum_{r=1}^{R-1} \ell(B_{r+1}) = \sum_{r=2}^R \ell(B_r)$, the terms $\ell(A_r), \ell(B_r)$ for $2 \leq r \leq R-1$ cancel on both sides, leaving

$$\ell(B_1) + \ell(A_R) = \ell(A_1) + \ell(B_R).$$

Since $B_R = 0$ and $A_1 = V$, $B_1 = U$, this reads

$$\ell(U) + \ell(A_R) = \ell(V).$$

As $\ell(A_R) \geq 0$, we conclude

$$\ell_{T_q}(U_q) \leq \ell_{T_q}(V_q).$$

□

AI comment ALG-09. Action required — unresolved proof or drafting issue. Beyond ALG-04: the exhaustion sentence (“this iterated-preimage process can be carried out only finitely many times”) is not a proof. One preimage suffices, taken *before* localizing: if $z \in G^p$ and $z = wx_1$ with $w \in G^{p-r+1}$ (possible for r large, since the filtration is exhaustive), then the class of z vanishes in B_r . See comparison COMP-06. Replace the incomplete exhaustion sentence in the printed proof; this comment alone is not an accepted proof edit.

AI comment COMP-06. Action required — unresolved proof or drafting issue. Localize only the T -modules on the pages, not Q . Before localization, choose one preimage $y = zx_1$ and use exhaustiveness to put z in some filtration piece; then y dies on a sufficiently late page. After localization the increasing kernels stabilize because U_q has finite length. Telescoping the exact page sequences gives the stated inequality. This supplies the missing exhaustion argument identified in ALG-09. [Mathematical proof \(COMP-06, p. 4\)](#).

On the other hand, we can also establish a different inequality.

Lemma 6.4. *For every minimal prime $\mathfrak{q}$ in $\text{Supp}_T U$,*

$$\ell_{T_q}(U_q) > \ell_{T_q}(V_q).$$

Proof. By Gabber involutivity (Theorem 3.8) and Lemma 3.9, every minimal prime $\mathfrak{p}$ of $\text{Ann}_C M$ is involutive. It follows, that no $\mathfrak{p}$ contains x_1 : Assume $x_1 \in \mathfrak{p}$. Since $P \in \mathfrak{p}$, also the Poisson brackets

$$\{x_1, P\}, \quad \{x_1, \{x_1, P\}\}, \dots$$

lie in $\mathfrak{p}$. $\{x_1, -\}$ differentiates once with respect to ξ_1 , and because P is homogeneous of degree N and monic in ξ_1 , its N th derivative is the nonzero constant $N!$ - therefore the N th bracket would put $N!(-1)^N N!$ in $\mathfrak{p}$. Since $\mathfrak{p}$ is a proper prime ideal, no nonzero constant can be in $\mathfrak{p}$, so we must conclude $x_1 \notin \mathfrak{p}$. This result holds under localization at $T \setminus \mathfrak{q}$, writing $M' = M_q$ for the finite module over $C' = (T \setminus \mathfrak{q})^{-1}C$.

Consider the increasing chain of kernels $\ker(x_1 : M' \rightarrow M') \subseteq \ker(x_1^2 : M' \rightarrow M') \subseteq \dots$. Since M' is Noetherian, this chain stabilizes at some point m where $K := \ker(x_1^m) = \ker(x_1^{m+1}) = \dots$.

AI comment EDIT-02. Proposed change — author acceptance pending. The undefined command `\cots` is aliased to `\cdots` in the annotation preamble so this draft compiles. The kernel-chain argument is repeated immediately below; keep the fuller second version when resolving this comment.

Write $M' = M_q$, a finite module over $C' = (T \setminus \mathfrak{q})^{-1}C$; this is the same localization used throughout. As shown above, no minimal prime of $\text{Ann}_C(M)$ contains x_1 , and this persists after localizing at $T \setminus \mathfrak{q}$: no minimal prime of $\text{Ann}_{C'}(M')$ contains x_1 either, **since the annihilator of the finite module M commutes with localization**.

The chain $\ker(x_1) \subseteq \ker(x_1^2) \subseteq \dots$ of submodules of M' stabilizes, since M' is Noetherian over C' ; let $K := \ker(x_1^m : M' \rightarrow M')$ at a stage where

it stabilizes, so $x_1 K \subseteq K$ and $\ker(x_1^{m+1}) = \ker(x_1^m) = K$. Each quotient $\ker(x_1^{i+1})/\ker(x_1^i)$ maps to $V_q = \ker(x_1 : M' \rightarrow M')$ injectively via $v \mapsto x_1^i v$, so, as V_q has finite length, so does each quotient and hence K itself.

Two elementary facts about K :

- $x_1^{-1}K = K$: if $v \in K$ then $x_1 v \in K$ (as $x_1 K \subseteq K$); if $x_1 v \in K = \ker(x_1^m)$ then $x_1^{m+1}v = 0$, so $v \in \ker(x_1^{m+1}) = K$.
- $K \cap x_1 M' = x_1 K$: " $\supseteq$ " is trivial. For " $\subseteq$ ", if $w = x_1 v \in K$ for some $v \in M'$, then $v \in x_1^{-1}K = K$, so $w \in x_1 K$.

For any endomorphism T of a finite-length module N , additivity along $0 \rightarrow \ker T \rightarrow N \rightarrow TN \rightarrow 0$ and $0 \rightarrow TN \rightarrow N \rightarrow N/TN \rightarrow 0$ gives $\ell(\ker T) = \ell(N/TN)$. Applying this to $T = x_1|_K$: since $\ker(x_1) \subseteq K$, we have $\ker(x_1|_K) = \ker(x_1) = V_q$, so

$$\ell(K/x_1 K) = \ell(V_q).$$

The natural surjection $M'/x_1 M' \rightarrow (M'/K)/x_1(M'/K)$ has kernel $(K + x_1 M')/x_1 M' \cong K/(K \cap x_1 M') = K/x_1 K$, giving the exact sequence

$$0 \longrightarrow K/x_1 K \longrightarrow U_q \longrightarrow (M'/K)/x_1(M'/K) \longrightarrow 0,$$

so by additivity and using $\ell(K/x_1 K) = \ell(V_q)$,

$$\ell(U_q) = \ell(V_q) + \ell((M'/K)/x_1(M'/K)).$$

It remains to show the last term is positive.

Since $q \in \text{Supp}_T U$, $U_q = M'/x_1 M' \neq 0$; choose $\mathfrak{P} \in \text{Supp}_{C'}(M'/x_1 M')$ and a minimal prime $\mathfrak{p} \subseteq \mathfrak{P}$ of M' . We have shown above that $x_1 \notin \mathfrak{p}$, so x_1 is a unit in the localization at $\mathfrak{p}$; as K is annihilated by a power of x_1 , $K_{\mathfrak{p}} = 0$. From $0 \rightarrow K_{\mathfrak{p}} \rightarrow M'_{\mathfrak{p}} \rightarrow (M'/K)_{\mathfrak{p}} \rightarrow 0$ we get $(M'/K)_{\mathfrak{p}} \cong M'_{\mathfrak{p}} \neq 0$, so $\mathfrak{p} \in \text{Supp}(M'/K)$; since support is closed under passing to larger primes and $\mathfrak{p} \subseteq \mathfrak{P}$, also $\mathfrak{P} \in \text{Supp}(M'/K)$, i.e. $(M'/K)_{\mathfrak{P}} \neq 0$.

Since $\mathfrak{P} \in \text{Supp}_{C'}(M'/x_1 M')$ and $M'/x_1 M'$ is annihilated by x_1 , we have $x_1 \in \mathfrak{P}$. If $x_1(M'/K)_{\mathfrak{P}} = (M'/K)_{\mathfrak{P}}$, then as x_1 lies in the maximal ideal of the local ring $C'_{\mathfrak{P}}$, Nakayama's lemma [23](see [38, Tag 00DV]) would force $(M'/K)_{\mathfrak{P}} = 0$, a contradiction. Hence $((M'/K)/x_1(M'/K))_{\mathfrak{P}} \neq 0$, so $(M'/K)/x_1(M'/K) \neq 0$.

Therefore $\ell(U_q) = \ell(V_q) + \ell((M'/K)/x_1(M'/K)) > \ell(V_q)$. $\square$

AI comment COMP-07. Information — proof comparison; no edit requested. The formal proof uses Gabber directly for minimal primes. If such a prime contained x_1 , the N -fold bracket of the monic symbol P with x_1 would put the unit $\pm N!$ in that prime. The subsequent stable-kernel length computation and Nakayama argument agree with the printed proof. [Mathematical proof \(COMP-07, p. 4\)](#).

Lemma 6.5. *The cokernel $U = M/x_1 M$ of right multiplication by x_1 on $M = \text{gr } Q$ vanishes, i.e. $U = 0$.*

Proof. If $U \neq 0$, it has a minimal support prime $\mathfrak{q}$. Lemma 6.2 makes the relevant lengths finite, and Lemmas 6.3 and 6.4 contradict one another. Thus $U = M/x_1 M = 0$. $\square$

7. CHARACTERISTICS OF THE CHARACTERISTIC VARIETY

Corollary 7.1. *Equip Q with the order filtration $Q_m := qF_m = \text{im}(F_m \rightarrow Q)$ induced by the cyclic generator q denoting the class of 1 in Q . Then*

$$(7.1) \quad \text{Char}(Q) \subseteq V(P) \subseteq T^*\mathbb{A}^n.$$

Proof. $Q_\bullet$ is a good filtration because Q is cyclic. Since $q\tilde{d} = [\tilde{d}] = 0$ in Q , the symbol $P = \sigma_N(\tilde{d})$ annihilates the class $\bar{q} \in Q_0/Q_{-1}$ of q in $\text{gr } Q$: the element $q\tilde{d} \in Q_N$ is zero, and its image in Q_N/Q_{N-1} is $\bar{q} \cdot P$. Since $\text{gr } Q$ is generated over $k_0[x, \xi]$ by $\bar{q}$, we get $P \in \text{Ann}_{k[x, \xi]} \text{gr } Q$, and $\text{Char}(Q) = V(\text{Ann } \text{gr } Q) \subseteq V(P)$ because V reverses inclusions. $\square$

Proposition 7.2. *The canonical quotient satisfies*

$$(7.2) \quad \text{Char}(Q) \cap \pi^{-1}(H) = \emptyset,$$

where $\pi : T^*\mathbb{A}^n \rightarrow \mathbb{A}^n$ is the cotangent projection; *equivalently, $\text{Ann}_C \text{gr } Q + (x_1) = C$, i.e. no prime of $\text{Supp}_C \text{gr } Q$ contains x_1 .*

Proof. By Lemma 6.5 $U = M/x_1 M = 0$. Since M is finite over C , its quotient by x_1 has support $\text{Supp}_C(M) \cap V(x_1)$. Lastly, $\text{Supp}_C(M) = \text{Char}(Q)$, and $V(x_1)$ is the set of all points $\in k^{2n}$ where $x_1 = 0$, which is nothing else than the pre-image of $H = \{x_1 = 0\}$ under the projection, i.e. $V(x_1) = \pi^{-1}(H)$. Since $M/x_1 M = 0$, it has no support, so $\text{Supp}(M/x_1 M) = \text{Supp}_C(M) \cap V(x_1) = \text{Char}(Q) \cap \pi^{-1}(H) = \emptyset$.

Here points are prime ideals of C : the argument is $\text{Supp}_C(M/x_1 M) = \text{Supp}_C M \cap V(x_1)$ for the finite C -module M . $\square$

Formal comparison: stated for the support transported along $\xi \mapsto -\xi$, which fixes x_1 ; transported back verbatim

AI comment COMP-08. Information — proof comparison; no edit requested. The formal conclusion is scheme-theoretic: $\text{Supp}(\text{gr } Q) \cap V(x_1) = \emptyset$, equivalently $\text{Ann}(\text{gr } Q) + (x_1) = (1)$. It therefore persists under field extension. The formal statement uses the support transported by $\xi \mapsto -\xi$; this fixes x_1 , so the avoidance statements are equivalent. [Mathematical proof \(COMP-08, p. 5\)](#).

8. A CONORMAL DIRECTION AT INFINITY

Throughout this section and the next, $\mathbb{C}$ may be replaced by any algebraically closed field K of characteristic zero; the formal statements are proved in that generality, and Section 10 uses $K = \bar{k}$. Let $X = \mathbb{A}_{\mathbb{C}}^n$ be the affine n -space over $\mathbb{C}$. Because $\mathbb{A}_{\mathbb{C}}^n$ is flat has a global coordinate frame, $T^*X = X \times (\mathbb{C}^n)^*$ holds and we can therefore identify all cotangent fibres with $(\mathbb{C}^n)^*$. We write $[Z_0 : \cdots : Z_n]$ for homogeneous coordinates on the

projective space $\mathbb{P}^n$, and embed $X \hookrightarrow \mathbb{P}^n$ as the affine chart $\{Z_0 \neq 0\}$ via $[1 : x_1 : \dots : x_n]$, so that $x_i = Z_i/Z_0$ and the hyperplane at infinity is $\{Z_0 = 0\}$. Let $Y \subset X$ be an irreducible closed subvariety of X . We define $\text{Dir}(Y)$ to be the Zariski closure in $\mathbb{P}((\mathbb{C}^n)^*)$ of the nonzero conormal vectors at all smooth affine points of Y projected onto $\mathbb{P}((\mathbb{C}^n)^*)$.

Formal comparison: Y given by a prime ideal; conormal space = annihilator of the Zariski tangent space; closure by homogeneous polynomials

Lemma 8.1. *Suppose a formal arc in the projective closure of Y has generic point in the smooth affine locus, and put $m = \dim Y$. Let $L \subset \mathbb{C}[[t]]^{n+1}$ be a direct summand of rank $m + 1$ whose generic fibre is the affine cone over the embedded projective tangent m -plane. If the reduction L_0 satisfies $L_0 \subset \{Z_1 = 0\}$, then $[dx_1] \in \text{Dir}(Y)$.*

Proof. To prove this lemma, we will construct an arc such that its generic fibre is a conormal direction at a smooth point of Y (hence lies in $\text{Dir}(Y)$), identify its special fibre as $[dx_1]$ and conclude from the closedness of $\text{Dir}(Y)$ that it must also lie in $\text{Dir}(Y)$. The annihilator $L^\perp$ of L is also a direct summand, and therefore reduces onto $L_0^\perp$. The dual basis vector $Z_1^\vee$ annihilates L_0 since $L_0 \subset \{Z_1 = 0\} = \ker(Z_1^\vee)$, and hence $Z_1^\vee \in L_0^\perp$. Because the reduction is a surjective map, we can lift $Z_1^\vee \in L_0^\perp$ to $\ell(t) \in L^\perp$, satisfying $\ell(0) = Z_1^\vee$. At the generic point, $\ell(t)$ annihilates the generic fibre of L and hence by hypothesis the cone over the embedded projective tangent space. The fibre coordinates $(\ell_1(t), \dots, \ell_n(t))$ of $\ell(t)$ in the affine chart $Z_0 \neq 0$ therefore form a nonzero affine conormal covector. Because $\ell(0) = Z_1^\vee \neq 0$, the reduction of the fibre coordinates at $t = 0$ is nonzero, so their projective direction extends over $\mathbb{C}[[t]]$. Specifically, this reduction is $(1, 0, \dots, 0)$, giving the special fibre $[dx_1]$. ~~Since $\text{Dir}(Y) \subset \mathbb{P}((\mathbb{C}^n)^*)$ is closed, hence proper over $\mathbb{C}$, $[dx_1]$ must lie in $\text{Dir}(Y)$ by the valuative criterion [17, Theorem II.4.7].~~ The generic direction lies in $\text{Dir}(Y)$: every homogeneous polynomial vanishing on $\text{Dir}(Y)$ vanishes on the conormal directions of the $\mathbb{C}((t))$ -point $\gamma(\eta)$ as well (the conormal variety over Y_{sm} is defined over $\mathbb{C}$, and its $\mathbb{C}$ -points are dense in it). Hence the defining equations of the closed set $\text{Dir}(Y)$ vanish on the direction of $\ell(t)$ in $\mathbb{C}((t))$, hence on its value at $t = 0$, which is $[dx_1]$. No valuative criterion is needed. $\square$

AI comment GEO-04. Proposed change — author acceptance pending. The printed proof did not justify that a conormal direction at the $\mathbb{C}((t))$ -point $\gamma(\eta)$ belongs to $\text{Dir}(Y)$, which is defined from $\mathbb{C}$ -points; the red text supplies the scalar-extension step, which the formalization proves as `scalarExtension_vanishing`.

Formal comparison: any algebraically closed field and any axis; auxiliary, not used by the formal proof of Theorem 8.2

AI addition: The generic conormal direction and the passage to $t = 0$.

Let η be the generic point of the arc, and write $\gamma(\eta) = [v_0 : \cdots : v_n]$ with $v_i \in \mathbb{C}((t))$ and $v_0 \neq 0$, since the generic point is affine. Put $y = (v_i/v_0)_{i \geq 1} \in Y(\mathbb{C}((t)))$ and let $T_y Y$ be the Zariski tangent space of $I(Y) \otimes \mathbb{C}((t))$ at y .

(1) *Dehomogenization.* For $\tau \in T_y Y$, the vector $(0, v_0 \tau)$ lies in the affine cone over the projective tangent plane, which is the generic fibre of L . Since $\ell(t)$ annihilates that fibre, $0 = v_0 \sum_{i \geq 1} \ell_i \tau_i$. Hence $\ell' = (\ell_1, \dots, \ell_n)$ annihilates $T_y Y$, and so $\ell' \in (T_y Y)^\perp = \text{span}\{\nabla f(y) : f \in I(Y)\}$, i.e. $(y, \ell'(t))$ is a $\mathbb{C}((t))$ -point of the equation-conormal locus $\mathcal{C}_{\mathbb{C}((t))}$ obtained by extending the equation-conormal locus of Y to $\mathbb{C}((t))$. Smoothness of y is not used.

(2) *Passage to $\mathbb{C}$ -points.* $\text{Dir}(Y)$ is defined through conormal vectors at $\mathbb{C}$ -points, and y is not one. Let P be a homogeneous polynomial vanishing at every direction in $\text{Dir}(Y)$. Then $\xi_1 P$ vanishes on $N(Y) = \bigcup_{z \in Y_{\text{sm}}} N_z^* Y$. By generic smoothness, this vanishing extends to the equation-conormal locus: a polynomial in gradient covectors vanishing on the dense smooth locus vanishes on all of Y . Parametrize covectors by finite linear combinations of gradients and apply the Nullstellensatz in $\mathbb{C}[x, \lambda]$; the vanishing then persists under extension to $\mathbb{C}((t))$. Thus $\ell_1(t)P(\ell'(t)) = 0$. Since $\ell_1(0) = 1$, ℓ_1 is a unit of $\mathbb{C}[[t]]$, hence $P(\ell'(t)) = 0$ in $\mathbb{C}[[t]]$.

(3) *Closedness replaces the valuative criterion.* The coefficients of P are constants and $\ell'(t) \in \mathbb{C}[[t]]^n$, so the constant term of $P(\ell'(t))$ is $P(\ell'(0)) = P(1, 0, \dots, 0)$, and it vanishes. As P was an arbitrary homogeneous equation of the closed set $\text{Dir}(Y)$, we get $[dx_1] \in \text{Dir}(Y)$. No properness is needed, because the limit point $\ell'(0) = e_1$ is given explicitly.

Formal comparison: generic point only Zariski-tangent; no smoothness needed for this step

AI comment COMP-10. Information — proof comparison; no edit requested. The formal tangent-limit criterion is auxiliary. It assumes a generically smooth arc and a direct-summand lattice whose generic span is the lifted tangent space, with the specified axis component vanishing on its residue. Its proof does not use the smoothness hypothesis or the stated rank, but these remain assumptions of the formal statement. The formal proof of Theorem 8.2 follows a separate Laurent-series route, so it does not certify every step of the printed tangent-limit argument. [Mathematical proof \(COMP-10, p. 5\)](#). See ROUTE-01 for the separate author decision on the printed geometric proof.

Theorem 8.2. *If Y is nonempty and $Y \cap H = \emptyset$, then*

$$[dx_1] \in \text{Dir}(Y).$$

Formal comparison: over every algebraically closed field of characteristic zero; the formal proof follows a different route, written out after the printed proof

AI comment ROUTE-01. Max — whole proof correspondence; author decision pending. Compare the printed geometric proof below with the checked statements, including its intermediate claims. Preserve the paper’s argument where a checked variant or adapter can support it. COMP-11 gives a checked Laurent-series proof if a genuine gap requires a replacement; a different Lean implementation alone is no reason to rewrite the paper. Johanna need only review a concrete text correction once its mathematical need is established.

Proof. If Y is zero-dimensional, its tangent space is trivial, so every cotangent direction occurs in its conormal fibre. If $\dim Y > 0$, but the coordinate function $x_1|_Y$ is constant on Y , it is a nonzero constant (since $Y \cap \{x_1 = 0\} = \emptyset$) and dx_1 annihilates every tangent space and is therefore a conormal vector to Y , i.e. $[dx_1] \in \text{Dir}(Y)$. We may therefore assume that $m = \dim Y > 0$ and that $x_1|_Y$ is nonconstant.

Let $\bar{Y} \subset \mathbb{P}^n$ be the closure of Y in $\mathbb{P}^n$ and let $\nu : \tilde{Y} \rightarrow \bar{Y}$ be its normalization. Normalization is finite in this setting [38, Tag 0BXR, Lemma 33.27.1]. The rational function $x_1 = Z_1/Z_0$ has a zero prime divisor D on $\tilde{Y}$, i.e. a prime divisor with $\text{ord}_D(x_1) > 0$. Otherwise x_1^{-1} would have no pole at any codimension-one point of $\tilde{Y}$. Since $\tilde{Y}$ is normal, x_1^{-1} would be globally regular [38, Tag 031T]. As $\tilde{Y}$ is finite (hence proper) over $\bar{Y}$, it is proper over $\mathbb{C}$. Finite normalization and preservation of projectivity are standard [38, Tag 0BXR, 0GK4, Lemmas 33.27.1–2]. By [38, Tag 04L2, Lemma 33.26.2], any globally regular function must be constant. This contradicts our standing assumption that $x_1|_Y$ is nonconstant. Since $Y \cap \{x_1 = 0\} = \emptyset$, x_1 does not vanish on Y is a unit on Y (Nullstellensatz), so the generic point of D lies outside $\nu^{-1}(Y)$, hence in the hyperplane $\{Z_0 = 0\}$ at infinity. We choose Z_j nonzero at the generic point of D , and define $q_i = Z_i/Z_j$. With respect to the divisorial valuation at D , we define

$$(8.1) \quad a = \text{ord}_D(q_0), \quad b = \text{ord}_D(q_1).$$

Indeed, because the generic point lies in $\{Z_0 = 0\}$, a must be positive; as $a, b > 0$ while $q_j = 1$, the index j differs from 0 and 1. b is finite since Z_1 does not vanish identically on $\bar{Y}$, and we recall $x_1 = Z_1/Z_0 = q_1/q_0$, therefore $\text{ord}_D(x_1) = \text{ord}_D(q_1) - \text{ord}_D(q_0) = b - a$. Because D was specifically chosen to be a zero prime divisor of x_1 , $\text{ord}_D(x_1) > 0$, hence $b > a$. We shrink a neighbourhood U of the generic point of D so that Z_j is invertible. Defining $D \cap U = (s)$, we can find regular functions w_0 and w_1 that do not vanish identically on D , such that

$$q_0 = s^a w_0, \quad q_1 = s^b w_1,$$

Restricting the normalization to D inherits finiteness, so at the generic point of D , the map has finite fibers. Because $\nu|_D$ is finite, its function-field extension is separable since $\mathbb{C}$ is a field of characteristic 0 ([38, Tag 0322]), and consequently, $\nu|_D$ is smooth at the generic point of D [38, Tag 07ND], applied to the dominant finite morphism $D \rightarrow \nu(D)$ onto its image. Since

smoothness is an open condition, $\nu|_D$ is smooth on a dense open subset of D . Equivalently, since D and $\nu(D)$ have equal dimension $m - 1$, its differential has rank $m - 1$.

AI comment GEO-01. Proposed change — author acceptance pending.
 The generic-smoothness step is valid for the dominant finite map $D \rightarrow \nu(D)$ (Tag 07ND). Name that image as the target and choose the point also over its smooth locus; then the differential into the projective chart has the asserted rank $m - 1$.

Choose a closed point p of $D \cap U$ in the smooth loci of both D and $\tilde{Y}$, where the differential of $\nu|_D$ has rank $m - 1$, and outside $V(w_0 w_1)$, i.e. $w_0 \neq 0$ and $w_1 \neq 0$ at the chosen point, **and lying over the smooth locus of $\nu(D)$** . Normality of $\tilde{Y}$ ensures that its singular locus does not contain D and because all conditions hold on a dense open subset of D , their finite intersection is nonempty, so such a point exists.

We can define a completed local ring at p , which is isomorphic to a power series ring [38, Tag 0C0S, Lemma 10.160.10]. We can choose a formal coordinate system centered around p with parameters $(t, z_2, \dots, z_m)$. We choose $t = s$: since our point lies in the smooth locus of $D = \{s = 0\}$, $ds \neq 0$ there, so s extends to a regular system of parameters.

The set $\tilde{Y} \setminus \nu^{-1}(Y_{\text{sm}})$ (with Y_{sm} being the smooth locus of Y) is a proper closed subset of $\tilde{Y}$ containing p . Let $I \subset \hat{\mathcal{O}}_{\tilde{Y}, p} \cong \mathbb{C}[[t, z_2, \dots, z_m]]$ be the ideal of this closed set in the completed local ring, so that every element of I vanishes at every point of this closed set, and choose any nonzero $f \in I$. We expand f in homogeneous parts,

$$f = f_d(t, z_2, \dots, z_m) + f_{d+1}(t, z_2, \dots, z_m) + \dots,$$

where $f_d \neq 0$ is the lowest-degree nonzero homogeneous part. Since $f_d(1, z_2, \dots, z_m)$ is a nonzero polynomial in $z_2, \dots, z_m$, it vanishes only on a proper closed subset of $\mathbb{C}^{m-1}$. We now choose constants $c_2, \dots, c_m$ outside this subset, so that

$$f_d(1, c_2, \dots, c_m) \neq 0.$$

When we change the coordinate system by replacing z_α by $z_\alpha - c_\alpha t$, we therefore guarantee that $f \neq 0$ for $t \neq 0$ along the (tilted) formal t -axis. Since f vanishes on all of $\tilde{Y} \setminus \nu^{-1}(Y_{\text{sm}})$ (as f lies in its ideal), this forces every point of the arc with $t \neq 0$ – in particular its generic point – to lie outside $\tilde{Y} \setminus \nu^{-1}(Y_{\text{sm}})$, i.e. in $\nu^{-1}(Y_{\text{sm}})$.

Setting $z_2 = \dots = z_m = 0$ in these coordinates therefore defines a transverse map $\gamma : \text{Spec } \mathbb{C}[[t]] \rightarrow \tilde{Y}$ whose generic point lies, by construction, in Y_{sm} , the smooth affine locus of Y .

In the new coordinates of the local ring, we can accordingly rewrite

$$(8.2) \quad q_0 = t^a u_0, \quad q_1 = t^b u_1,$$

where u_0, u_1 are units since $w_0(p) \neq 0, w_1(p) \neq 0$.

In the coordinates $(t, z_2, \dots, z_m)$, each q_i is a regular function on $\tilde{Y}$ near p , so $v = (q_0, \dots, q_n)^\top$, with $q_j = 1$ is a local parametrization, in these coordinates, of ν 's image in the chart $Z_j \neq 0$. We define the partial derivative of v with respect to t and the z_α

$$V_\alpha = \frac{\partial v}{\partial z_\alpha}, \quad T = \frac{\partial v}{\partial t}.$$

V_α vanish in rows 0, 1, and j at $t = 0$, since $q_j = 1$ is constant, and q_0 and q_1 (and hence their derivatives) have factors t vanishing at $t = 0$. Since locally, $D = \{t = 0\}$, they are the images of a tangent basis of D under the differential of $\nu|_D$. Because we chose p such that $\nu|_D$ would have full rank and is hence injective, the images V_α of the tangent basis of D are also independent. Since the reductions of $V_2, \dots, V_m$ are independent and vanish in rows 0, 1, j , they have rank $m - 1$, so there exist $m - 1$ further coordinate rows $i_1, \dots, i_{m-1}$ (necessarily distinct from 0, 1, j) in which the corresponding $(m - 1) \times (m - 1)$ submatrix of $(V_2, \dots, V_m)$ is invertible. Because this minor is invertible at $t = 0$, it remains invertible over $\mathbb{C}[[t]]$, so there are unique coefficients $\lambda_2(t), \dots, \lambda_m(t) \in \mathbb{C}[[t]]$ solving

$$\sum_{\alpha=2}^m \lambda_\alpha(t) (V_\alpha)_{i_k} = T_{i_k}, \quad k = 1, \dots, m - 1.$$

Setting

$$T' := T - \sum_{\alpha=2}^m \lambda_\alpha(t) V_\alpha$$

therefore gives a column vector whose entries in rows $i_1, \dots, i_{m-1}$ vanish identically by construction. The entries of V_α in rows 0 and 1 have valuation at least a and b respectively, whereas

$$\partial_t q_0 = at^{a-1}u_0 + O(t^a), \quad \partial_t q_1 = bt^{b-1}u_1 + O(t^b).$$

Because we work over a field of characteristic zero, the displayed leading coefficients are nonzero, so

$$(8.3) \quad \text{ord}_t(T_0) = a - 1, \quad \text{ord}_t(T_1) = b - 1.$$

Since the entries of V_α in rows 0, 1 have strictly higher valuation, subtracting a combination of $V_2, \dots, V_m$ from T to form T' cannot change these orders, hence

$$(8.4) \quad \text{ord}_t(T'_0) = a - 1, \quad \text{ord}_t(T'_1) = b - 1.$$

Let c be the minimum valuation of the entries of T' . Then $c \leq a - 1$, and $\widehat{T} = t^{-c}T'$ is regular and its evaluation at $t = 0$ is nonzero. Moreover

$$(8.5) \quad \text{ord}_t(\widehat{T}_1) \geq b - a > 0.$$

Since γ 's generic point lies in Y_{sm} , and normalization is an isomorphism over the smooth locus, **by the finite birational normal-target criterion**[38, Tag 0AB1], $\tilde{Y}$ and Y agree there; working over $\mathbb{C}((t))$, we may therefore

regard v as a local parametrization of Y itself near γ 's generic point. Since $(t, z_2, \dots, z_m)$ is a full coordinate system on the m -dimensional $\tilde{Y}$, the m vectors $T, V_2, \dots, V_m$ form a tangent basis at that point; together with v itself, they span the affine cone over the projective tangent m -plane to Y along γ . Since T' differs from T only by a combination of $V_2, \dots, V_m$, and $\hat{T}$ differs from T' only by multiplication by the invertible scalar $t^{-c} \in \mathbb{C}((t))^\times$, replacing T by T' and then by $\hat{T}$ does not change this span.

Also evaluated at $t = 0$, the columns $v, V_2, \dots, V_m, \hat{T}$ are independent vectors. Indeed, $v_j = 1$, while the other vectors evaluated at $t = 0$ have j th coordinate zero. $V_\alpha(0)$ were chosen to be independent among themselves, and by construction $\hat{T}(0) \neq 0$. ~~If $\hat{T}(0)$ were a combination of $v(0), V_2(0), \dots, V_m(0)$, matching the zero entries in rows $i_1, \dots, i_{m-1}$ would force the $V_\alpha(0)$ -coefficients to vanish (by invertibility of the minor), and matching row j would then force the $v(0)$ -coefficient to vanish too (since $v_j(0) = 1 \neq 0 = \hat{T}_j(0)$), contradicting that $\hat{T}(0)$ is nonzero, hence also $\hat{T}(0)$ is independent from the other vectors.~~ ~~If $\hat{T}(0)$ were a combination of $v(0), V_2(0), \dots, V_m(0)$, matching row j first forces the $v(0)$ -coefficient to vanish (since $v_j(0) = 1$ while all other vectors have j th entry 0); then matching the rows $i_1, \dots, i_{m-1}$, where $\hat{T}(0)$ vanishes, forces the $V_\alpha(0)$ -coefficients to vanish by invertibility of the minor. So $\hat{T}(0) = 0$, a contradiction.~~ $v, \hat{T}, V_2, \dots, V_m$ therefore span the direct-summand tangent lattice L required by Lemma 8.1 and their evaluations at $t = 0$ span the reduction L_0 . As a remark, we note that L_0 is the Grassmannian limit.

AI comment GEO-02. Proposed change — author acceptance pending. In the independence argument above, use row j first: it forces the coefficient of $v(0)$ to be zero. Only then do the selected minor rows force all $V_\alpha(0)$ coefficients to vanish. Otherwise the selected rows may contain a contribution from $v(0)$. Nonzero $\hat{T}(0)$ gives the contradiction and a unit maximal minor gives the direct-summand lattice.

All vectors $v, \hat{T}, V_2, \dots, V_m$ evaluated at $t = 0$ have zero Z_1 -coordinate, so

$$L_0 \subset \{Z_1 = 0\}.$$

By Lemma 8.1 $[dx_1] \in \text{Dir}(Y)$. □

AI comment COMP-11. Information — proof comparison; no edit requested. The formal proof works over every algebraically closed field of characteristic zero. It replaces the tangent-limit route by a Laurent-series argument. The constant-coordinate case is treated separately; otherwise a divisorial valuation gives $dx_1 = \sum_j c_j dx_j$ with each c_j of positive valuation. The conormal covector $e_1 - \sum_j c_j e_j$ therefore tends to e_1 , and the equation-conormal closure and scalar-extension arguments give the required direction. The detailed valuation and residue argument is in the technical supplement; it does not justify a gap in a different printed proof. [Mathematical proof \(COMP-11, p. 6\)](#). See ROUTE-01 for the separate author decision on the printed geometric proof.

Formal comparison: any algebraically closed field of characteristic 0; (i) is the affine-cone form used in Section 9

9. THE INVOLUTIVE OBSTRUCTION

Definition 9.1 (Fibre-conical). *A closed subset $W \subseteq T^*X$ is fibre-conical if it is stable under the fibrewise scaling $(y, \xi) \mapsto (y, t\xi)$, $t \in \mathbb{G}_m$.*

Theorem 9.2 (Involutive exclusion). *Let $W \subset T^*X$ be a reduced closed fibre-conical and involutive subset. Let $P \in \mathbb{C}[\xi_1, \dots, \xi_n]$ be homogeneous and suppose $P(1, 0, \dots, 0) \neq 0$. There is no nonempty W satisfying*

$$W \cap \pi^{-1}(H) = \emptyset, \quad W \subset V(P).$$

AI addition: Formal statement of Theorem 9.2.

Lean proves the following strengthening. Let K be an algebraically closed field of characteristic zero and $n \geq 1$. Let $W \subset K^n \times K^n$ be nonempty and Zariski closed, with $(y, a\xi) \in W$ whenever $(y, \xi) \in W$ and $a \in K^\times$, and with $\{I(W), I(W)\} \subseteq I(W)$. Let $P \in K[\xi_1, \dots, \xi_n]$ be any polynomial, not necessarily homogeneous, with $P(\xi) = 0$ for all $(y, \xi) \in W$ and $P(1, 0, \dots, 0) \neq 0$. Then some $(y, \xi) \in W$ has $y_1 = 0$, i.e. $W \cap \pi^{-1}(H) \neq \emptyset$. The proof uses only the base-relative fragment $\{f \circ \pi, I(W)\} \subseteq I(W)$ for $f \circ \pi \in I(W)$. The Lean bracket is the negative of the one in the paper, which does not affect involutivity. “Involutive” means $\{I(W), I(W)\} \subseteq I(W)$, not that $I(W)$ is a Poisson ideal. For $K = \mathbb{C}$ this contains Theorem 9.2.

Formal comparison: any algebraically closed field of characteristic 0; P need not be homogeneous

Proof. Supposing that such a W exists, we define $Y = \overline{\pi(W)}$. We write $I(Y)$ and $I(W)$ for the radical vanishing ideals, and define $W_y = W \cap T_y^*X$ (using the identification $T_y^*X \cong (\mathbb{C}^n)^*$ as before). Closedness and fibre-conicality imply

$$(9.1) \quad Y \times \{0\} \subset W.$$

Every nonempty conical fibre contains its origin, so in fact $\pi(W) = \{y : (y, 0) \in W\}$ is closed and equals Y . In particular, $Y \cap H = \emptyset$.

For $f \in I(Y)$, its pullback belongs to $I(W)$. Involutivity gives $\{f, I(W)\} \subset I(W)$. The Hamiltonian derivation of f is

$$H_f = \{-, f\} = \sum_i \frac{\partial f}{\partial x_i} \frac{\partial}{\partial \xi_i}.$$

Due to the involutivity, H_f preserves W . Since H_f involves only $\partial/\partial \xi_i$, it strictly lowers ξ -degree, hence is locally nilpotent on $\mathbb{C}[x, \xi]$; its exponential is therefore a well-defined, finite sum on any polynomial, equal to vertical translation by $t df$: for any point $(y, \xi) \in T^*X$, solving the flow equations $\dot{y} = 0$, $\dot{\xi} = \partial_y f$ gives $(y, \xi) \mapsto (y, \xi + t df_y)$. These translations commute, so starting from $(y, 0) \in W$ (eq. (9.1)) and composing them for varying

$f \in I(Y)$ and $t \in \mathbb{C}$ shows that every point $(y, \sum_i t_i d(f_i)_y)$ lies in W , i.e. $\text{span}\{df_y : f \in I(Y)\} \subset W_y$.

Let Y_0 be an irreducible component of Y . At a general smooth point y of Y_0 outside the other components, $I(Y)_y = I(Y_0)_y$, so at this point, $\text{span}\{df_y : f \in I(Y)\} = \text{span}\{df_y : f \in I(Y_0)\} = N_y^*Y_0$, the conormal space to Y_0 . Since $N_y^*Y_0 \subset W_y$ holds at a dense open set of points in the smooth locus $y \in (Y_0)_{\text{sm}}$ and W is closed, the closure of the conormal bundle over the smooth locus $T_{Y_0}^*X := \overline{\bigcup_{y \in (Y_0)_{\text{sm}}} N_y^*Y_0}$ lies in W

$$(9.2) \quad T_{Y_0}^*X \subset W.$$

Since Y_0 is an irreducible component of the closed set Y , Y_0 is nonempty, irreducible and itself closed. Furthermore, from $Y \cap H = \emptyset$ follows $Y_0 \cap H = \emptyset$, so the ~~hypotheses~~ ~~hypotheses~~ of Theorem 8.2 are fulfilled and hence $[dx_1] \in \text{Dir}(Y_0)$. Since $\text{Dir}(Y_0)$ is, by definition, the projectivization of $T_{Y_0}^*X$ (the closure of the conormal directions over the smooth locus of Y_0), this places $[dx_1]$ in the closure of the projective fibre directions of $T_{Y_0}^*X$, hence of W by (9.2). Since P is homogeneous in the fibre variables ξ alone, the condition $P(\xi) = 0$ depends only on the direction $[\xi]$, not on the base point x or the scale of ξ ; it therefore defines a well-defined closed hypersurface $V_+(P) \subset \mathbb{P}((\mathbb{C}^n)^*)$, the projectivized vanishing locus of P . As $W \subset V(P)$, every projectivized fibre direction of W – regardless of its base point, even one escaping to infinity – lies in this same $V_+(P)$. ~~Since $[dx_1]$ is a projectivized fibre direction of W , $[dx_1] \in V_+(P)$.~~ Every conormal direction at a smooth point of Y_0 is a fibre direction of W and hence lies in the closed set $V_+(P)$; therefore so does their closure $\text{Dir}(Y_0) \ni [dx_1]$. Thus $[dx_1] \in V_+(P)$, or equivalently $P(1, 0, \dots, 0) = 0$, a contradiction to the theorem's hypotheses. Therefore, no such W that is nonempty can exist.

AI comment GEO-03. Proposed change — author acceptance pending. The preceding result places $[dx_1]$ in the *closure* of projective conormal directions, not necessarily in a fibre of W at a finite base point. Every conormal direction lies in $V_+(P)$; that set is closed, so the limiting direction does too. This yields the same contradiction $P(1, 0, \dots, 0) = 0$.

□

AI comment COMP-13. Information — proof comparison; no edit requested. The formal proof first excludes x_1 from every minimal base prime by iterated Poisson brackets with P . Assuming avoidance of H , it then forces those primes to be homogeneous in the transverse base coordinates and uses an asymptotic conormal direction to contradict $P(1, 0, \dots, 0) \neq 0$. This is an alternative route, not a verification of each intermediate step above. [Mathematical proof \(COMP-13, p. 8\)](#). See ROUTE-01 for the separate author decision on the printed geometric proof.

AI comment GEO-05. Optional refinement — not required. Optional statement refinements suggested by the formal version, for the authors to decide: in Section 8 say that Y is reduced and irreducible (so $I(Y)$ is prime) and identify $(\mathbb{C}^n)^*$ with $\mathbb{C}^n$ via dx_i ; in Lemma 8.1 the arc is $\mathbb{C}[[t]]$ -valued with affine generic point, and “smooth” can be weakened to “Zariski tangent space”; in Theorem 8.2 “ Y nonempty” is automatic, and the affine-cone form (i) of the red proof is what Theorem 9.2 actually uses; in Theorem 9.2 “reduced” is vacuous for a point set, the last sentence should read “if $W \neq \emptyset$ and $W \subset V(P)$, then $W \cap \pi^{-1}(H) \neq \emptyset$ ”, and homogeneity of P can be dropped.

10. PROOF OF THEOREM 1.1

AI comment ALG-05. Proposed change — author acceptance pending. Updated 2026-09-30: the red text below now works over $\bar{k}$ as the formalization does, states the base-change lemma and the fibre-conicality, and replaces the placeholder citations. For descent, flat field extension gives $\mathrm{gr}(Q \otimes_{k_0} \mathbb{C}) \cong (\mathrm{gr} Q) \otimes_{k_0} \mathbb{C}$. Empty support of the finite graded module implies it is zero; exhaustiveness and $F_{-1}Q = 0$ then imply $Q_{\mathbb{C}} = 0$. Faithful flatness detects $Q = 0$ [38, Tag 00HP]. The placeholder citations below are retained visibly as proposed removals.

Let $K = \bar{k}$ and $Q_K := Q \otimes_k K = A_n(K)/(\tilde{d}A_n(K) + x_1^N \tilde{d}A_n(K))$ (right exactness of $\otimes$). The order filtration of Q_K is the base change of that of Q , and the order initial ideal of the extended right ideal is the extension of the initial ideal: $\mathrm{Ann}_{K[x, \xi]} \mathrm{gr} Q_K = (\mathrm{Ann}_{k[x, \xi]} \mathrm{gr} Q) K[x, \xi]$ (a k -basis of $\mathrm{gr} Q$ in each degree stays a K -basis; this is the lemma `filteredInitialLifting` of the formalization). Consequently $\mathrm{Char}(Q_K) \subseteq V(P)$ and, since $\mathrm{Ann} \mathrm{gr} Q + (x_1)$ is the unit ideal by Proposition 7.2, $\mathrm{Char}(Q_K) \cap \pi^{-1}(H) = \emptyset$. Alternatively, Sections 4–7 apply verbatim to $\tilde{d} \in A_n(K)$.

Over $\mathbb{C}K$, we define W as the reduced characteristic variety of Q

$$W = \mathrm{Char}(Q_{\mathbb{C}K})_{\mathrm{red}}$$

with $Q_{\mathbb{C}K} := Q \otimes_{k_0} \mathbb{C}K$. ~~Under the right-to-left equivalence used above,~~ $(x, \xi) \mapsto (x, -\xi)$ carries W to the characteristic variety of the corresponding ~~left module~~. Gabber’s involutivity theorem [15, 16] (Theorem 3.8, applied to the right ideal of $A_n(K)$ generated by $\tilde{d}, x_1^N \tilde{d}$) therefore makes W involutive: its vanishing ideal is $\sqrt{\mathrm{Ann} \mathrm{gr} Q_K}$ by the Nullstellensatz; see also [19, Theorem 2.3.1 and Appendix D] and [30]. ~~Closedness and fibre-conicality are preserved, as is containment in $V(P)$, because P is homogeneous of degree N and so $P(-\xi) = (-1)^N P(\xi)$.~~ W is closed, and it is fibre-conical because $\mathrm{Ann} \mathrm{gr} Q_K$ is homogeneous for the grading by ξ -degree ($\mathrm{gr} Q_K$ is a graded module), so $V(\mathrm{Ann} \mathrm{gr} Q_K)$ is stable under $\xi \mapsto t\xi$. By eq. (7.1) and the base-change remark, $W \subset V(P)$, by eq. (4.3) $P(1, 0, \dots, 0) = 1 \neq 0$, and by eq. (7.2), $W \cap \pi^{-1}(H) = \emptyset$. By Theorem 9.2 W must therefore be empty.

A variety is empty, iff its reduction is empty, since reducing does not change the underlying point-set, so $\mathrm{Char}(Q_{\mathbb{C}K}) = \emptyset$. A nonzero **finitely generated** module with an exhaustive filtration satisfying $F_{-1} = 0$ has nonzero associated

graded module, and a nonzero finite module over $K[x, \xi]$ has ~~coherent filtered module has~~ nonempty ~~associated-graded~~ support (every proper ideal lies in a maximal ideal). ~~\cite{somewhere}~~ Consequently $\text{Char}(Q_{\underline{\mathbb{C}}K}) = \emptyset$ implies $Q_{\underline{\mathbb{C}}K} = 0$. Since field extensions are always faithfully flat ~~\cite{somewhere}~~ (a k -vector space V with $V \otimes_k K = 0$ is zero), $\underline{\mathbb{C}}K$ is a faithfully flat module over k_0k , and therefore Q vanishes also over k_0k [38, Tag 00HP]

$$Q_{\underline{\mathbb{C}}K} = 0 \Leftrightarrow Q = 0.$$

By eq. (5.1), it follows that

$$A_n(k_0) = \tilde{d}A_n(k_0) + x^N \tilde{d}A_n(k_0)$$

or equivalently, there exist $R, S \in A_n(k_0)$ with

$$1 = \tilde{d}R + x^N \tilde{d}S.$$

~~This identity of course extends to the original field $k \ni k_0$.~~ Lastly, we can apply the inverse of the automorphism Φ_g , defined in Remark 4.3. This results in the final expression

$$1 = d(c^{-1}\Phi_g^{-1}(R)) + F d(c^{-1}\Phi_g^{-1}(S)) \quad F = \Phi_g^{-1}(\underline{x}x_1)^N,$$

proving Theorem 1.1.

AI addition: Base change of the order initial ideal.

Lemma 10.1. *Let K/k be a field extension, $I = \tilde{d}A_n(k) + x_1^N \tilde{d}A_n(k)$ and $I_K = \tilde{d}A_n(K) + x_1^N \tilde{d}A_n(K)$. Then $\text{in}(I_K) = \text{in}(I) \cdot K[x, \xi]$. Consequently $\text{Char}(Q_K)$ is the preimage of $\text{Char}(Q)$ under $\text{Spec } K[x, \xi] \rightarrow \text{Spec } k[x, \xi]$. Moreover $Q = 0 \iff \text{in}(I) = (1) \iff \text{in}(I_{\bar{k}}) = (1)$.*

Proof. Identify $A_n(K) = A_n(k) \otimes_k K$ through the PBW basis. Then $F_L(A_n(K)) = F_L \otimes_k K$ and $I_K = I \otimes_k K$ as K -subspaces. Since K is flat over k , $(I \otimes_k K) \cap (F_L \otimes_k K) = (I \cap F_L) \otimes_k K$. Hence every $z \in I_K \cap F_L(A_n(K))$ is a sum $\sum_i c_i w_i$ with $c_i \in K$ and $w_i \in I \cap F_L$. Its order- L symbol is $\sum_i c_i \sigma_L(w_i) \in \text{in}(I)K[x, \xi]$. The reverse inclusion holds because σ_L commutes with the scalar extension.

For the last claim: if $\text{in}(I) = (1)$, then $\text{gr } Q = 0$. Since $F_{-1}Q = 0$ and the filtration is exhaustive, induction on m gives $F_m Q = 0$ for all m , so $Q = 0$. The converse is clear. Finally, a proper ideal of $k[x, \xi]$ has a zero in $\bar{k}^{2n}$ (Nullstellensatz), so it stays proper in $\bar{k}[x, \xi]$. $\square$

Formal comparison: stated for the canonical ideal; the argument uses nothing else

AI comment COMP-15. Information — proof comparison; no edit requested. The formal main proof runs the geometry over $\bar{k}$ and descends the order initial ideal using Lemma 10.1; it does not use the embedding into $\mathbb{C}$. The rank-zero and scalar cases are handled separately. For the remaining cases, emptiness of characteristic support gives $1 \in I$, and the inverse symplectic change returns the fixed-source identity over k . Mathematical proof (COMP-15, p. 9).

~~Latly~~Lastly, as Stafford already showed, from $1 = dR + FdS$ follows cyclicity of the torsion right module:

Corollary 10.2. *Every finitely generated torsion right $A_n(k)$ -module is cyclic.*

AI comment ALG-06. Proposed change — author acceptance pending. State the consequence for *finitely generated* torsion right modules, as required by Stafford's two-generator theorem. Write $\phi(r, s) = t_1r + t_2s$. The map ϕ is surjective, not an isomorphism. Correct the final quotient to $T \cong (A_n \oplus A_n)/M$. The displayed right-module generator calculation itself is valid.

Proof. Suppose for every $d \neq 0 \in A_n$, there exists $F, R, S \in A_n$, such that $1 = dR + FdS$. Let us denote a torsion module by T . ~~Stafford's theorem (Theorem 3.7 [31, p. 438]) established, that every finitely generated torsion module can be generated by 2 generators,~~ Stafford's theorem [31, Theorem 3.7, p. 438], stated there for left modules and valid for right modules by the left-right symmetry noted on [31, p. 437], shows that every finitely generated torsion right A_n -module is two-generated (the Lean proof below avoids this theorem), so let us denote the generators of T with t_1 and t_2 . We can define a homomorphism

$$\phi : A_n \oplus A_n \rightarrow T \quad \phi(r, s) \mapsto t_1r + t_2s.$$

This ~~isomorphism~~map is surjective, since t_1, t_2 generate all of T . By the first isomorphism theorem (for $\phi : M \rightarrow N$, $\text{im}(\phi) \cong M/\ker(\phi)$), it follows

$$T \cong (A_n \oplus A_n)/M$$

with $M = \ker(\phi)$.

T is torsion and therefore $\forall t \in T : \exists m \in A_n \setminus \{0\} : tm = 0$. Specifically, let us denote with d_1 and d_2 the annihilators of t_1 and t_2 , i.e.

$$t_1d_1 = t_2d_2 = 0.$$

A_n is a right Ore domain, which means it has the property that for any nonzero d_1, d_2 there exists elements $r_1, r_2 \in A_n$ such that

$$d_1r_1 = d_2r_2 \neq 0.$$

Defining therefore $d := d_1r_1 = d_2r_2$ yields us a common annihilator of t_1 and t_2 .

$(d, 0)$ and $(0, d)$ are elements in $M = \ker(\phi)$, since $\phi(d, 0) = t_1d = 0$ and $\phi(0, d) = t_2d = 0$.

Because $(0, d), (d, 0) \in M$, also $(d, 0)S + (0, d)R = (dS, dR) = (dS, 1 - FdS) =: (a, 1 - Fa) \in M$, i.e. $(a, 1 - Fa)$ is an element of M .

On the other hand, $A_n \oplus A_n$ can be generated by $(a, 1 - Fa)$ and $(1, -F)$. We can check this by finding u, v such that for all $(x, y) \in A_n \oplus A_n$ we can write

$$(x, y) = (a, 1 - Fa)u + (1, -F)v.$$

Suitable choices are $u = y + Fx$ and $v = x - a(y + Fx)$.

Lastly, since $A_n \oplus A_n$ can be generated by $(a, 1 - Fa)$ and $(1, -F)$ and

$(a, 1 - Fa)$ is an element of M , it follows that $T \cong \underline{A_n \oplus A_n} / M(A_n \oplus A_n) / M$ can be generated by the image of $(1, -F)$, i.e. T is cyclic. $\square$

AI comment COMP-16. Information — proof comparison; no edit requested. The formal pair argument now exposes exactly the generator $t_1 - t_2 F$ used above. Its finite-generation induction avoids Stafford's two-generator theorem and the Ore step; this implementation difference does not require changing the printed generator calculation. Stafford defines “stably m -generated” on [31, p. 435]: every generating list with $r + 1$ entries, $r \geq m$, can be shortened to $a_i + a_{r+1} f_i$ ($1 \leq i \leq r$). Thus “stably cyclic” means $m = 1$ and is stronger than ordinary cyclicity in general. The corollary here asserts ordinary cyclicity, exactly as its proof shows. Mathematical proof (COMP-16, p. 10).

11. OUTLOOK

~~We have proved~~ The cofactors R, S are obtained non-constructively: the argument proves $Q = 0$ without producing a certificate. Effective bounds for $\deg_B R$ and $\deg_B S$ in terms of $\deg_B d$, and an algorithm in the spirit of the constructive two-generator methods, remain open. The Global Stafford extension raises the same questions for rings of differential operators on smooth affine varieties.

AI comment EDIT-01. Action required — unresolved proof or drafting issue. This sentence and the Outlook section are unfinished. Please complete the intended outlook or remove this fragment once reviewed.

REFERENCES

1. Christopher Albert, *AlgebraicAnalysis v0.3.0*, Zenodo software release v0.3.0, 2026, DOI: <https://doi.org/10.5281/zenodo.22666517>.
2. ———, *AlgebraicAnalysis v0.3.2*, Zenodo software release v0.3.2, 2026, DOI: <https://doi.org/10.5281/zenodo.22710109>.
3. ———, *Global Stafford*, Palomar Registry, <https://palomar-registry.org/entry?id=PALOMAR-2026-09-09-000001&version=2>, 2026, PALOMAR-2026-09-09-000001, version 2.
4. ———, *Global Stafford: Lean 4 formalization v1.0.6*, Zenodo software release v1.0.6, 2026, DOI: <https://doi.org/10.5281/zenodo.22669444>.
5. ———, *Stafford 3.8*, Palomar Registry, <https://palomar-registry.org/entry?id=PALOMAR-2026-09-05-000007&version=2>, 2026, PALOMAR-2026-09-05-000007, version 2.
6. ———, *Stafford's Conjecture 3.8 for Weyl algebras: Lean 4 formalization v1.2.0*, Zenodo software release v1.2.0, 2026, DOI: <https://doi.org/10.5281/zenodo.23072839>.
7. Gwyn Bellamy, *Module structure of Weyl algebras*, Journal of the London Mathematical Society **113** (2026), no. 1, e70373.
8. Jan-Erik Björk, *Rings of differential operators*, North-Holland Mathematical Library, vol. 21, North-Holland, Amsterdam, 1979.
9. Napoleon Caro and Daniel Levcovitz, *On a theorem of Stafford*, Cadernos de Matemática **11** (2010), 63–70, Preprint at <https://arxiv.org/abs/1005.4427>; journal PDF at the authors' university repository.
10. Napoleon Caro-Tuesta and Daniel Levcovitz, *Module structure of certain rings of differential operators*, Algebras and Representation Theory **23** (2020), 1637–1657.
11. Frédéric Chyzak, Alban Quadrat, and Daniel Robertz, *OreModules: a symbolic package for the study of multidimensional linear systems*, Applications of Time Delay Systems, Lecture Notes in Control and Information Sciences, vol. 352, Springer, 2007, pp. 233–264.
12. P. M. Cohn, *Algebra*, 2nd ed., vol. 1, John Wiley & Sons, Chichester, 1982.
13. Severino C Coutinho, *A primer of algebraic d-modules*, no. 33, Cambridge University Press, 1995.
14. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28 (Cham), Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
15. Ofer Gabber, *The integrability of the characteristic variety*, American Journal of Mathematics **103** (1981), no. 3, 445–468.
16. ———, *The integrability of the characteristic variety: erratum*, American Journal of Mathematics **104** (1982), no. 5, 1125.
17. Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer, New York, 1977.
18. Andre Hillebrand and Wiland Schmale, *Towards an effective version of a theorem of Stafford*, Journal of Symbolic Computation **32** (2001), no. 6, 699–716.
19. Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki, *D-modules, perverse sheaves, and representation theory*, Progress in Mathematics, vol. 236, Birkhäuser, Boston, 2008.
20. LeanProver contributors, *Comparator: statement and proof comparison for Lean*, Project-pinned commit 575674928e239f5bc452aab72d1dd7b0f1326494. Source: <https://github.com/leanprover/comparator/tree/575674928e239f5bc452aab72d1dd7b0f1326494>.
21. Anton Leykin, *Algorithmic proofs of two theorems of Stafford*, Journal of Symbolic Computation **38** (2004), no. 6, 1535–1550.

22. Hideyuki Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics, vol. 8, Cambridge University Press, Cambridge, 1986, Translated from the Japanese by M. Reid.
23. Tadasi Nakayama, *A remark on finitely generated modules*, Nagoya Mathematical Journal **3** (1951), 139–140.
24. NanoDa: *an independent kernel checker for Lean*, Project-pinned commit 68d5ca9db226849b41a6fff59d796ff19d0a8840. Source: https://github.com/ammkr/nanoda_lib/tree/68d5ca9db226849b41a6fff59d796ff19d0a8840.
25. Øystein Ore, *Theory of non-commutative polynomials*, Annals of Mathematics **34** (1933), no. 3, 480–508.
26. Palomar Registry, *About the Palomar registry*, Official documentation, accessed 2026-09-30: <https://palomar-registry.org/about.html>.
27. Alban Quadrat and Daniel Robertz, *Computation of bases of free modules over the Weyl algebras*, Journal of Symbolic Computation **42** (2007), no. 11–12, 1113–1141.
28. ———, *A constructive study of the module structure of rings of partial differential operators*, Acta Applicandae Mathematicae **133** (2014), no. 1, 187–234.
29. Christian Schnell, *D-modules*, 2019, Lecture notes.
30. Jyoti Singh and Shiv Datt Kumar, *On the involutivity of the characteristic variety*, Communications in Algebra **42** (2014), no. 8, 3607–3618.
31. J. T. Stafford, *Module structure of Weyl algebras*, Journal of the London Mathematical Society. Second Series **18** (1978), no. 3, 429–442.
32. ———, *Generating modules efficiently: algebraic K-theory for noncommutative noetherian rings*, Journal of Algebra **69** (1981), no. 2, 312–346.
33. ———, *On the stable range of right noetherian rings*, Bulletin of the London Mathematical Society **13** (1981), no. 1, 39–41.
34. ———, *Generating modules efficiently over noncommutative rings*, Séminaire d'Algèbre Paul Dubreil et Marie-Paule Malliavin, Proceedings. Paris 1981 (34ème Année), Lecture Notes in Mathematics, vol. 924, Springer, Berlin, 1982, pp. 72–88.
35. ———, *Corrigendum*, Journal of Algebra **82** (1983), no. 1, 294–296, Corrigendum to Stafford's 1981 article on generating modules efficiently.
36. ———, *Stably free, projective right ideals*, Compositio Mathematica **54** (1985), no. 1, 63–78, Available at https://www.numdam.org/item/CM_1985__54_1_63_0/.
37. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (New York), ACM, 2020, pp. 367–381.
38. The Stacks Project Authors, *Stacks project*, 2026.

APPENDIX A. AI-ASSISTED RESEARCH AND FORMALIZATION

Proof discovery, counterexample searches, review, and formalization was assisted by agentic AI tools. The work began with cyclic-extension computations in Macaulay2 and Maple. Attempts to construct universal cofactors directly produced special cases and reductions but eventually stalled. The decisive change was to study the canonical quotient $Q = A_n/(dA_n + x^N dA_n)$: Euler surjectivity and monicity constrained its characteristic support, while Gabber's involutivity and asymptotic conormal geometry forced that support to be empty. This turned cofactor selection into a quotient-vanishing problem. Full Lean formalization was deferred until the paper argument was complete, although small formal checks continued throughout. In the final formal development, support avoidance over the coordinate hyperplane was proved by a localized filtered Koszul length contradiction.

The workflow separated proposed ideas from verified claims. One controller maintained the authoritative proof state, while parallel workers explored distinct mechanisms and returned evidence with an explicit scope. Independent reviewers checked fixed revisions for quantifiers, sidedness, circularity, and missing hypotheses. The project protocols were based on a `math-frontier` agent skill developed for this purpose: freeze the claim, identify its first unresolved dependency, and specify how a route could fail. Conceptual prompts requested a small number of distinct mechanisms; paper-only tasks sought complete arguments before implementation. A condensed representative prompt, adapted from the archived protocols, is:

Work on one precisely stated unresolved claim. Propose at most three distinct mechanisms, each with an intermediate lemma, a noncircular justification, and an adversarial test. Preserve the quantifiers and right-sided cofactors. For a paper task, supply a complete argument; for formalization, prove the required hypotheses without adding axioms or using `sorry`. Return evidence and the first unsupported bridge; do not promote a conditional result to a theorem.

Research and review sessions used GPT-5.5, GPT-5.6 Luna and Sol, GPT-6 Astra, and Claude Fable 5, Haiku 4.5 and Opus 5. The project protocols assigned routine implementation and replay to Luna and conceptual consultation to Sol; independent reports also record Opus reviews. Subsequent manuscript, proof-map, build and release work used Fable 5.1, Opus 5, Sonnet 5 and GPT-6 Luna. These records establish participation and task roles, without assigning a particular mathematical idea to one model.

The completed proof uses Lean 4 and Mathlib and pins AlgebraicAnalysis v0.3.0 [14, 37, 1]; the later library archive is v0.3.2 [2]. Formalization distinguished theorems proving required inputs from conditional theorems merely consuming them. Statement comparison and independent NanoDa and Lean kernel checks verified the submitted declarations [20, 24, 26]; their proofs use only `propext`, `Classical.choice`, and `Quot.sound`. The Stafford38 and Global Stafford headline proofs are registered separately in Palomar [5, 3]; their Lean developments are archived [6, 4].

APPENDIX B. FURTHER WORK

AI comment REF-01. Proposed change — author acceptance pending. The original appendix’s undefined labels `prop:symbol` and `eq:charinP` are visibly replaced below with the existing order-symbol and characteristic-containment labels.

~~AI also proved some related stuff regarding Right Ore localizations, the corresponding left-hand identity, and apparently the corresponding identity over other rings such as differential rings with laurent series coefficients; I am leaving that out for now.~~ The formalization also proves the following consequences, which are not otherwise used here: the identity in every right Ore localization of $A_n(k)$; the left-handed identity $1 = Rd + SdF$ via

the formal adjoint ; the identity for finite-order differential operators on localizations of $k[x_1, \dots, x_n]$, including principal opens, Laurent-polynomial coefficients and the fraction field ; and, independently of the geometric argument, an evolutionary version with arbitrary coefficients commuting with one Weyl pair .

AI comment SCOPE-01. Proposed change — author acceptance pending.

Review the scope before accepting this paragraph. The coefficients in the stated localization result are Laurent polynomials, not arbitrary Laurent series, and the operators have finite order. Retain further consequences only with their exact hypotheses and proof references; the declaration index is in the technical supplement.

B.1. Noncharacteristic H. An apparently nice ~~(though I do not understand its value)~~ side theorem that comes from "restriction to the coordinate hyperplane" (after [Corollary 7.1](#)), is that $H = \{x_1 = 0\}$ is noncharacteristic; noncharacteristic restriction is what makes the pull-back of Q to H well behaved ([Cauchy–Kovalevskaya–Kashiwara](#)):

The nonzero conormal covectors of H are the multiples λdx (or in coordinates $(\lambda, 0, \dots, 0)$), with $\lambda \neq 0$. Since P is homogeneous and $P(1, 0, \dots, 0) = 1$, we have $P(\lambda, 0, \dots, 0) = \lambda^N \neq 0$, therefore $\lambda dx \notin V(P)$. Since $\text{Char}(Q) \subseteq V(P)$ by [\(7.1\)](#), $\text{Char}(Q)$ therefore contains no nonzero conormal covector to H . More precisely, the projection

$$V(P) \cap T^*\mathbb{A}^n|_H \longrightarrow T^*H$$

is finite: on coordinate rings it adjoins ξ_1 subject to the monic equation $P(\xi_1, \dots, \xi_n) = 0$. Its restriction to $\text{Char}(Q)$ is therefore finite, which is the noncharacteristic condition [\[29, Definition 15.6\]](#).

Definition B.1 (Noncharacteristic, [\[29, Def. 15.6\]](#)). *Let $i: H \hookrightarrow X$ be a smooth closed subvariety and M a coherent $\mathcal{D}_X$ -module. Let $\varpi: T^*X|_H \rightarrow T^*H$ be the canonical surjection. Then H is noncharacteristic for M if the restriction $\varpi|_{\text{Char}(M) \cap T^*X|_H}: \text{Char}(M) \cap T^*X|_H \rightarrow T^*H$ is a finite morphism.*

Formal comparison: for $X = \mathbb{A}^n$, $H = \{x_1 = 0\}$ and a characteristic ideal J : the coordinate-ring map $k[T^*H] \rightarrow k[x, \xi]/(J + (x_1))$ is finite

Proposition B.2 ($H = \{x = 0\}$ is noncharacteristic for Q). *With P as in [\(4.2\)](#) and $H = \{x_1 = 0\}$, the projection $V(P) \cap T^*X|_H \rightarrow T^*H$ is finite. Hence H is noncharacteristic for Q . Equivalently, $\text{Char}(Q) \cap T_H^*X$ is contained in the zero section.*

Proof. Write $x' = (x_2, \dots, x_n)$, $\xi' = (\xi_2, \dots, \xi_n)$, so that $T^*X|_H = \text{Spec } k[x', \xi_1, \xi']$ and ϖ forgets ξ_1 . By [Proposition 4.5](#), P is monic of degree N in ξ_1 with coefficients in $k[\xi']$; therefore

$$k[x', \xi'] \longrightarrow k[x', \xi_1, \xi']/(P)$$

is a finite ring extension, generated by $1, \xi_1, \dots, \xi_1^{N-1}$. Thus $V(P) \cap T^*X|_H \rightarrow T^*H$ is finite, and by [\(7.1\)](#) so is its restriction to

$\text{Char}(Q) \cap T^*X|_H$. For the last statement, the conormal space is $T_H^*X = \{x_1 = 0, \xi' = 0\}$, on which $P(\xi_1, 0, \dots, 0) = \xi_1^N$; so $V(P) \cap T_H^*X = \{\xi = 0\}$ is the zero section. $\square$

Formal comparison: the zero-section statement holds without $x_1 = 0$: tangential covariables vanishing forces $\xi_1 = 0$

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